PD-Prune: Disaggregated Prefill-Decode Pruning for Efficient Agentic LLM Serving

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Abstract

Agentic and inference-time search workloads often expand a committed context into multiple candidate continuations, many of which are later discarded. We focus on branch-select workloads in which one continuation is eventually committed as the next spine. Under prefill–decode (PD) disaggregation, each admitted candidate may incur residual prefill and KV-transfer cost before decode-side pruning takes effect. We propose PD-Prune, a cross-stage pruning framework that moves branch filtering to the earliest stage with sufficient information. On the prefill side, PD-Prune applies pre-kernel admission using KV residency and a lightweight model-free heuristic score, so rejected residuals consume neither residual prefill nor connector bandwidth. Candidates with uncertain prefill-side scores can be refined by a short decode-side probe, while existing decode-time pruning methods operate unchanged on the admitted set. A pruning-aware copy-on-write context tree manages shared trunk state, reclaims aborted residual pages, and retains the committed spine for the next step. On Qwen3-8B, prefill-side admission reduces computed prefill by 23% at matched committed-answer accuracy; on MATH-500, at $k{=}16$ a 128-token probe increases coverage to $20/32$ at $0.69\times$ the wall-clock of the full fan-out; on GSM8K with Qwen3-14B, retaining only the committed spine reduces median peak KV usage to $0.72\times$ that of automatic prefix caching (APC) at matched committed-answer accuracy.

1 Introduction

Agentic LLM applications interleave generation with branching. A coding agent proposes several candidate edits or tool calls for the same repository context, executes one, and discards the rest [36, 34, 30]; an inference-time search controller expands a Tree-of-Thoughts frontier or a self-consistency pool from one partial solution [35, 31, 32]. Specifically, at each step the serving system starts from a committed context, the *trunk* x_u, and expands it into k candidate continuations. We call the candidate-specific suffix ρ_i a *residual*, so candidate i is the sequence $x_u \| \rho_i$. We focus on branch-select workloads in which one continuation is eventually committed as the *spine* and becomes the trunk of the next step, while the remaining branches are discarded [8, 12, 17]. This pattern requires the system to serve a fan-out of candidates even though only one lineage continues.

Branching improves search coverage, but it also multiplies serving cost. A residual that must be expanded further processes any still-uncached input (*prefill*) and then generates tokens (*decode*). Production stacks increasingly place these phases on separate instances under prefill–decode (PD) disaggregation, so the KV pages produced by prefill must also cross a connector before decode can begin [42, 25]. Most residuals are eventually discarded, yet before they are pruned they may already have consumed prefill compute, KV transfer, and some decode. As the agent repeats this fan-out over many steps, the cost of non-surviving candidates accumulates. Stopping an unnecessary residual earlier is therefore a first-class opportunity for efficient PD-disaggregated agentic serving.

Prior work improves efficiency mainly along two axes: path pruning and KV-cache optimization. Decode-time path pruners cut low-value branches after partial generation: ESC stops sampling once a window agrees [17], Speculative Rejection halts low-reward prefixes at a horizon [26], DPTS early-stops children after a mini-step [10], and STOP predicts futility from a short prefix [4]. These methods depend on generated text, so by the time they prune, residual prefill and the connector transfer have often already been paid and cannot be recovered. KV systems attack redundant work on the shared trunk: automatic prefix caching (APC) in vLLM and RadixAttention in SGLang reuse resident trunk pages [14, 41], and agent sched-

ulers with KV retention manage context across tool calls [23, 16, 33, 9, 29]. Even when the trunk is fully shared, however, each residual's uncached suffix must still be computed and transferred—work that is wasted if that candidate is later discarded. APC neither rejects a residual nor controls which pages cross the connector. Combined with PD disaggregation, pruning therefore remains *stage-local*: each stage can discard only work that has already reached it. The gap is that existing methods reduce duplicate compute and later decode, yet leave the prefill and KV-transfer cost of low-value candidates largely untouched. We ask how to cut that early-stage cost in PD-disaggregated agent serving without sacrificing committed-answer accuracy.

Achieving this goal, however, presents two key challenges.

Challenge 1: Prefill-side admission. Before a residual runs, the signals that existing pruners rely on—generated tokens—are unavailable, and KV residency is split across instances: a page resident on the prefill side may still require transfer or recompute on the decode side. The controller must decide, cheaply and from pre-execution information alone, whether each residual should be skipped, prefilled, transferred, or decoded.

Challenge 2: Decode-side probing. A candidate judged low-value before prefill may still reach a correct answer after further generation. Aggressive early rejection saves resources but can hurt committed-answer accuracy; retaining every residual preserves accuracy but erodes the gain. When the pre-execution signal is inconclusive, a short subsequent generation should refine the decision before the residual consumes a full decode budget.

These decisions also allocate KV pages with different lifetimes. Siblings share the trunk but own distinct residual pages, so an aborted branch must free its private pages, and the committed spine must stay resident as the trunk of the next agent step.

Disaggregated pruning. Our key insight is to disaggregate pruning across the prefill and decode stages. PD-Prune performs early admission before prefill when sufficient information is available, defers uncertain decisions to decode-side probing, and uses lightweight KV state management to support branch reclamation and spine retention (Figure 1). On the prefill side, a controller assigns each residual one of $\{\textsc{Hit}, \textsc{Admit}, \textsc{Probe}, \textsc{Reject}\}$ before any kernel runs, using dual-side page residency and a prefill-side heuristic score (§4.1). Residuals whose scores fall in an uncertainty band are probed: they are prefilled, decoded for t_0 tokens, and extended only if the generated prefix clears a decode-side threshold τ_{dec} calibrated separately from the prefill-side threshold τ_{pre} (§4.2). Existing decode-time pruners such as ESC, Speculative Rejection, and DPTS operate unchanged on the admitted branches. Lightweight KV state management supports these decisions: a copy-on-write context tree shares the trunk across siblings, transfers only decoder-missing pages, frees residual pages on abort, and retains the committed spine (§4.4).

Results. On Qwen3-8B GSM8K with $k{=}16$, prefill-side admission in front of ESC, Speculative Rejection, or DPTS reduces computed prefill from 5008 to 3848 tokens (-23%) for every decode-time pruner; committed-answer accuracy is unchanged, and ESC wall-clock falls from $16.49\,\mathrm{s}$ to $14.45\,\mathrm{s}$ (§5). On MATH-500, aggressive prefill-side rejection lowers any-finished coverage from $20/32$ to $13/32$, whereas at $k{=}16$ a 128-token probe increases coverage to $20/32$ at $72.2\,\mathrm{s}$ instead of $104.1\,\mathrm{s}$. The supporting state management, which retains only the committed spine, reduces median peak KV usage on the full GSM8K test set with Qwen3-14B to $0.72\times$ that of APC at matched committed-answer accuracy, and the ratio stays within 0.72–$0.75\times$ across six model families. All measurements run on physically disaggregated vLLM prefill and decode instances connected by a KV connector.

Contributions. Our contributions are summarized as follows:

- We formulate pruning in disaggregated agentic serving as a per-residual, cross-stage decision problem, and make explicit the cache regimes and residency conditions under which APC does or does not share the trunk (§3).
- We propose *disaggregated pruning* and instantiate it in PD-Prune with prefill-side admission and decode-side probing; lightweight KV state management supports branch reclamation and spine retention (§4).
- We demonstrate that prefill-side admission composes with three existing decode-time pruning methods and reduces computed prefill, and that decode-side probing mitigates the coverage loss

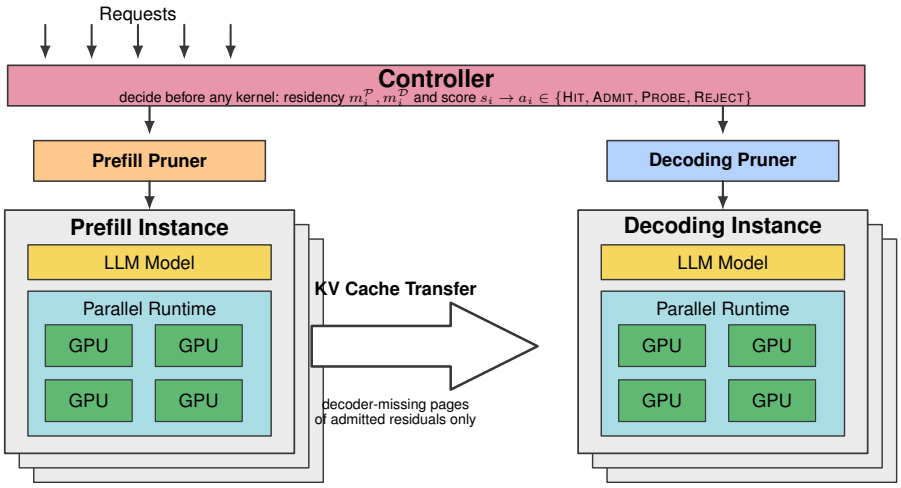


Figure 1: **PD-Prune runtime.** The controller reads the residency of each residual on the prefill instance ($m_i^{\mathcal{P}}$) and the decode instance ($m_i^{\mathcal{D}}$) together with a lightweight model-free heuristic score s_i (the prefill-side score), and assigns an action $a_i \in \{\text{HIT}, \text{ADMIT}, \text{PROBE}, \text{REJECT}\}$ before any kernel runs. The prefill instance computes only the non-resident suffix of admitted and probed residuals; the connector carries only the pages the decode instance lacks. The decode-side pruner extends a probed residual beyond t_0 tokens only if its generated prefix clears τ_{dec}. After commit, only the spine $x_\star$ is retained.

of aggressive prefill-side rejection, while preserving committed-answer accuracy (§5).

2 Related Work

Inference-time search and path pruning. Chain-of-thought elicits intermediate steps [32]; Tree-of-Thoughts, self-consistency, and Monte Carlo tree search expand many partial solutions and then select [35, 31, 39]. Accuracy and token cost both grow with the number of traces. Decode-time path pruners cut that cost after generation has started: ESC stops once a window agrees [17], Speculative Rejection halts low-reward prefixes at a horizon [26], DPTS early-stops children after a mini-step [10], and STOP organizes such signals by source and learnability and trains an internal futility token [4]. EquivPruner merges semantically equivalent steps so that textually distinct but mathematically identical actions are not explored twice [21]. All of these decide from generated or written text and are agnostic to where the KV cache was computed. PD-Prune is complementary: it adds a pre-kernel decision and a decode-side probe, while these decode-time pruners operate unchanged on the admitted set; a learned scorer such as STOP's could serve as $\text{Score}_{\text{dec}}$.

Prefix sharing and PD disaggregation. Orca schedules at iteration granularity [37]; vLLM pages the KV cache and hashes full blocks for APC [14]; SGLang shares prefixes through a radix tree [41]; Sarathi chunks prefills to co-schedule them with decode [2, 3]. DistServe and Splitwise place prefill and decode on separate instances and move KV across a connector [42, 25]. These mechanisms determine whether a trunk is shared and how pages move, but they serve every request they receive. Whether k siblings share the trunk depends on the cache regime (§3.2); even when the trunk is fully reused, each residual's uncached suffix is still computed and transferred. PD-Prune uses residency on both instances as an admission input and treats the connector as the channel whose volume admission reduces.

Agentic serving, retention, and copy-on-write. Autellix and Hermes schedule at program granularity [23, 22]; ThunderAgent and SAGA optimize workflows [13, 11]; Parrot, Ayo, and Pythia exploit application structure [20, 27, 38]; JITServe allocates bandwidth under heterogeneous SLOs [40]. InferCept, Continuum, and MORI retain KV across tool calls [1, 16, 33]; ConServe disaggregates prefill and decode scheduling at conversation granularity from observed agent behavior [9]; Sutradhara overlaps tool execution with the prefill of a single tool-independent prefix [5]. ForkKV is closest in mechanism: it shares a base KV cache copy-on-write across LoRA agents [29]. CachePrune edits attention state inside one request [28]; speculative decoding drafts *output* tokens rather than a future prompt prefix [15, 6, 18]. These systems place,

schedule, or retain contexts that will be served; ConServe in particular disaggregates scheduling but not pruning. PD-Prune targets the complementary decision of which sibling continuations should be served at all, and at which PD stage that decision is made. Its copy-on-write context tree is the state-management mechanism that supports those decisions for siblings of one agent step.

Appendix E expands the literature survey.

3 Problem Formulation

We model one agent step as a branching step served by a prefill instance $\mathcal{P}$ and a decode instance $\mathcal{D}$ connected by a KV connector. Table 7 in the appendix summarizes the notation.

Definition 1 (Branching step)**.** A parent u holds a committed trunk x_u with $|x_u|{=}L$. A fan-out is a set of candidate continuations. We call the candidate-specific suffix ρ_i a *residual*, with $|\rho_i|{=}\ell_i$; candidate i is the sequence $x_u\|\rho_i$. Index 0 denotes the highest-ranked candidate produced by the branching controller (e.g., the greedy continuation). Commit selects one continuation $\star$ and leaves the spine $x_\star = x_u\|\rho_\star$, which becomes the trunk of the next step.

With b bytes of KV per token, three occupancies characterize one step:

$$\begin{aligned} M_{\text{clone}} &= b\big(kL + \textstyle\sum_i \ell_i\big), \\ M_{\text{CoW}} &= b\big(L + \textstyle\sum_i \ell_i\big), \\ M_{\text{spine}} &= b\big(L + \ell_\star\big). \end{aligned} \tag{1}$$

M_{clone} arises only when every request materializes its own trunk; M_{CoW} holds one shared trunk and all residuals; M_{spine} is the occupancy after uncommitted residuals are freed.

3.1 Residency on Two Instances

Paged engines index full blocks of P tokens by a chained key $h_j = \text{hash}(h_{j-1}, \text{block}_j, \text{extra})$ [14, 41], so a block is matched only together with its entire parent prefix. For an instance $\mathcal{X} \in \{\mathcal{P}, \mathcal{D}\}$ we define the resident prefix of the trunk and of residual i as

$$\begin{aligned} c_u^{\mathcal{X}} &= \max\{jP \le L :\ x_u[0{:}jP) \sqsubseteq \mathcal{X}\}, \\ m_i^{\mathcal{X}} &= \max\{jP \le \ell_i :\ x_u\|\rho_i[0{:}jP) \sqsubseteq \mathcal{X}\}, \end{aligned} \tag{2}$$

where $z \sqsubseteq \mathcal{X}$ means that every page of z is physically resident on $\mathcal{X}$, and $m_i^{\mathcal{X}}{=}0$ whenever $c_u^{\mathcal{X}}{<}L$, because chained keys make a residual page unreachable without its full trunk. For exposition we take L to be a multiple of P; otherwise the partial trunk page is charged to each residual. Three distinctions matter. (i) *Parent-aware match:* $m_i^{\mathcal{X}}$ is defined on $x_u\|\rho_i$, not on ρ_i alone. (ii) *Index presence versus residency:* an index entry whose block has been evicted does not count; $m_i^{\mathcal{X}}$ counts physically resident pages only. (iii) *Instance locality:* pages resident on $\mathcal{P}$ are not visible on $\mathcal{D}$, so $m_i^{\mathcal{P}}$ and $m_i^{\mathcal{D}}$ generally differ.

3.2 Cache Regimes

Whether the trunk is shared across the k siblings depends on the cache state and on the engine, not on APC alone. Let W^{trunk} be the trunk tokens computed on $\mathcal{P}$ for one fan-out.

R1 *Cold, concurrent.* The trunk is not resident and the siblings are admitted before any of them publishes trunk blocks; depending on how the scheduler handles in-flight blocks, $L \le W^{\text{trunk}} \le kL$.

R2 *Cold, serialized.* The trunk is prefilled once (by a warm-up request or by deferring siblings until it is published); $W^{\text{trunk}}{=}L$ at the cost of one serialization step.

R3 *Warm.* The trunk is already resident on $\mathcal{P}$; $W^{\text{trunk}}{=}0$.

R4 *Explicit fork.* The branching controller declares the fan-out and the trunk is aliased copy-on-write; $W^{\text{trunk}}{=}L - c_u^{\mathcal{P}}$ without serialization.

APC attains R2 or R3 in many agentic deployments, so trunk sharing alone is not the source of the savings we report. All GPU measurements in §5 warm the trunk before the fan-out (R3); measured prefill reductions are therefore attributable to residual admission rather than to cross-step trunk synchronization. On the connector, the trunk crosses at most once, $X^{\text{trunk}}{=}L - c_u^{\mathcal{D}}$. After commit, the spine is the logical trunk of the next step; its KV residency on $\mathcal{P}$ and $\mathcal{D}$ is restored according to the underlying KV synchronization policy.

3.3 Objective

Each residual receives an action a_i from a set $\mathcal{A}$, inducing computed prefill W_i, connector tokens X_i, and decode tokens d_i; we write $W{=}\sum_i W_i$, $X{=}\sum_i X_i$, and $D^+{=}\sum_i d_i$. We formulate the design objective as

$$\begin{aligned} \min_{\pi}\ & \mathbb{E}\big[T_{\text{pre}}(W) + \mu X + T_{\text{dec}}(D^+)\big] \\ \text{s.t.}\ & \mathbb{P}[\text{committed answer correct}] \ge Q_{\min}, \\ & M_{\text{peak}} \le M_{\text{pool}}, \qquad a_0 \ne \textsc{Reject}, \end{aligned} \tag{3}$$

where μ is the per-token transfer time. The defining constraint is informational: a decision taken before prefill may depend only on $(c_u^{\mathcal{P}}, c_u^{\mathcal{D}}, m_i^{\mathcal{P}}, m_i^{\mathcal{D}})$ and the residual text, whereas a decision taken on $\mathcal{D}$ may additionally read generated tokens. Decode-time pruners act only at the second point; PD-Prune adds the first and coordinates the two. PD-Prune instantiates this objective with a lightweight threshold-based policy.

4 PD-Prune: Cross-Stage Pruning

PD-Prune is a cross-stage pruning framework. The *decision* is pre-kernel admission, taken on the controller before any kernel runs (§4.1); uncertain candidates are deferred to a short decode-side *probe* that refines the decision (§4.2); a pruning-aware copy-on-write context tree manages the KV *state* created by these decisions (§4.4). Algorithm 1 is the per-fan-out implementation; Appendix B.6 gives the multi-step pipeline. §4.3 shows how the three combine in prefill and transfer volume.

4.1 Pre-Kernel Admission

For each residual the controller observes $o_i = (m_i^{\mathcal{P}}, m_i^{\mathcal{D}}, s_i)$, where $m_i^{\mathcal{X}}$ is the parent-aware resident prefix of (2) and $s_i = \mathrm{Score}_{\mathrm{pre}}(\rho_i) \in [0,1]$ is a lightweight model-free heuristic score computed from the residual text in $O(\ell_i)$ CPU time. We refer to s_i as the prefill-side score. Our scorer combines repetition detection and syntactic checks (Appendix B.4); a learned scorer can replace it without changing the policy.

Definition 2 (Admission action)**.** Given thresholds $\tau_{\mathrm{pre}} \le \tau_{\mathrm{dec}}$, residual i receives the first matching action in

$$a_i = \begin{cases} \textsc{Hit} & m_i^{\mathcal{D}} = \ell_i, \\ \textsc{Admit} & i = 0 \text{ or } s_i \ge \tau_{\mathrm{dec}}, \\ \textsc{Reject} & s_i < \tau_{\mathrm{pre}}, \\ \textsc{Probe} & \text{otherwise.} \end{cases} \tag{4}$$

HITrequires residency on the *decode* instance, since a page resident only on $\mathcal{P}$ must still be transferred. Because $i{=}0$ is caught by the first two cases, $a_0 \in \{\textsc{Hit}, \textsc{Admit}\}$: the top-ranked residual can be neither rejected nor probed away. Residuals in the uncertainty band $[\tau_{\mathrm{pre}}, \tau_{\mathrm{dec}})$ are probed. Setting $\tau_{\mathrm{pre}}{=}\tau_{\mathrm{dec}}$ empties the band and yields aggressive prefill-side rejection ("Blanket"); replacing PROBEby ADMITyields *reject-only admission*. Let $e_i{=}[\mathrm{Score}_{\mathrm{dec}}(y_i) \ge \tau_{\mathrm{dec}}]$ indicate that the probe y_i of a probed residual is extended (§4.2). The action determines the per-residual costs

$$W_i = \begin{cases} 0 & \textsc{Hit}, \textsc{Reject}, \\ \ell_i - m_i^{\mathcal{P}} & \textsc{Admit}, \\ \ell_i - m_i^{\mathcal{P}} + e_i W_i^{\mathrm{ext}} & \textsc{Probe}, \end{cases} \tag{5}$$

$$X_i = (\ell_i - m_i^{\mathcal{D}})\,[a_i \in \{\textsc{Admit}, \textsc{Probe}\}], \tag{6}$$

$$d_i = \begin{cases} 0 & \textsc{Reject}, \\ D & \textsc{Hit}, \textsc{Admit}, \\ t_0 + e_i(D - t_0) & \textsc{Probe}, \end{cases} \tag{7}$$

where D is the decode budget, t_0 the probe length, and W_i^{ext} the tokens recomputed when a probe is extended (§4.2). The trunk is aliased (§4.4) and contributes $W^{\mathrm{trunk}}{=}L - c_u^{\mathcal{P}}$ and $X^{\mathrm{trunk}}{=}L - c_u^{\mathcal{D}}$ once per fan-out. A decode-time pruner (ESC, Speculative Rejection, DPTS) may further shorten d_i for any admitted or extended residual.

4.2 Decode-Side Probe

A probed residual is prefilled, transferred, and decoded for t_0 tokens; the decode-side pruner then scores the generated prefix, $s_i' = \mathrm{Score}_{\mathrm{dec}}(y_i)$, and extends the residual to D only if $s_i' \ge \tau_{\mathrm{dec}}$. The two thresholds read different text: τ_{pre} reads the residual before generation, and τ_{dec} reads model output. They are therefore calibrated separately. A useful pair satisfies the design condition

$$\sup_{\rho \in \mathcal{C}_{\mathrm{rej}}} s(\rho) < \tau_{\mathrm{pre}} \le \inf_{\rho \notin \mathcal{C}_{\mathrm{rej}}} s(\rho), \qquad \sup_{y \in \mathcal{C}_{\mathrm{stop}}} s'(y) < \tau_{\mathrm{dec}}, \tag{8}$$

where $\mathcal{C}_{\mathrm{rej}}$ is the class of residuals that should never be prefilled (here, repetitive residuals) and $\mathcal{C}_{\mathrm{stop}}$ the class of probes that should not be extended. Table 1 lists the scores of our heuristic scorers on the three residual classes of our workload; with $(\tau_{\mathrm{pre}}, \tau_{\mathrm{dec}}) = (0.15, 0.45)$ both conditions hold. Because the heuristics assign near-constant scores per class, this separation is specific to our workload; a learned scorer can replace them on natural agent traces (§6).

Extension cost. If the probe's pages remain resident, extension continues decoding without recomputation and $W_i^{\mathrm{ext}}{=}0$. Our prototype issues the extension as a continuation request; when the probe's pages have been evicted in the meantime, the engine recomputes them, so $0 \le W_i^{\mathrm{ext}} \le \ell_i + t_0$.

Table 1: Scores of the heuristic scorers on the three residual classes of the evaluation mix. $\mathrm{Score}_{\mathrm{pre}}$ reads the residual; $\mathrm{Score}_{\mathrm{dec}}$ reads a probe that has not yet produced (malformed) or has produced (valid) a finished answer marker.

	repetitive	malformed	valid
$\mathrm{Score}_{\mathrm{pre}}(\rho_i)$	0.02	0.22	1.00
$\mathrm{Score}_{\mathrm{dec}}(y_i)$	0.04	0.16	0.92

Algorithm 1 PD-Prune (one fan-out). Actions follow (4); costs follow (5)–(7).

Require: trunk x_u; residuals $\{\rho_i\}$; residency on $\mathcal{P}/\mathcal{D}$; $(\tau_{\mathrm{pre}}, \tau_{\mathrm{dec}}, t_0, D)$
Ensure: spine $x_\star$; (W, X, D^+)

Phase I: Admit ▷ decide before any kernel
$a_i \leftarrow \pi(m_i^{\mathcal{P}}, m_i^{\mathcal{D}}, \mathrm{Score}_{\mathrm{pre}}(\rho_i))$ for all i
$\mathcal{A} \leftarrow \{i : a_i \in \{\textsc{Admit}, \textsc{Probe}\}\}$; $\mathcal{U} \leftarrow \{i : a_i{=}\textsc{Probe}\}$
$\mathrm{Fork}(x_u, \mathcal{A})$ ▷ alias trunk; residual pages only

Phase II: Prefill & Ship ▷ pay only what $\mathcal{P}/\mathcal{D}$ miss
$\mathrm{Prefill}_{\mathcal{P}}(x_u, \{\rho_i\}_{i\in\mathcal{A}})$ ▷ $\mathcal{P}$-miss; trunk once
$\mathrm{Ship}(x_u, \{\rho_i\}_{i\in\mathcal{A}})$ ▷ $\mathcal{D}$-miss

Phase III: Probe & Commit ▷ refine on $\mathcal{D}$, keep the spine
for all $i \in \mathcal{U}$ **do**
  $y_i \leftarrow \mathrm{Dec}(x_u \| \rho_i;\ t_0)$
  if $\mathrm{Score}_{\mathrm{dec}}(y_i) < \tau_{\mathrm{dec}}$ **then**
    $\mathrm{Abort}(\rho_i)$; $\mathcal{A} \leftarrow \mathcal{A} \setminus \{i\}$
  end if
end for
$\mathcal{S} \leftarrow \{i : a_i{=}\textsc{Hit}\} \cup \mathcal{A}$
$D^+ \leftarrow \mathrm{Prune}(\mathcal{S}; D)$; $\mathrm{Commit}(x_\star)$; $\mathrm{Abort}(\mathcal{S} \setminus \{\star\})$
return $x_\star, W, X, D^+$

This term is negligible at $k \leq 8$ but substantial at $k{=}16$ in our runs (Appendix D.5), and pinning probe pages until the extension decision is an engineering fix we leave to future work.

4.3 Work and Transfer Accounting

We compare PD-Prune with a baseline that serves the same fan-out without admission.

Assumption 1 (Baseline)**.** The baseline admits every residual, computes $W_{\mathrm{B}}^{\mathrm{trunk}}$ trunk tokens according to its cache regime (§3.2), computes $\ell_i - m_i^{\mathcal{P}}$ tokens for residual i, and transfers at least the pages each request lacks on $\mathcal{D}$, i.e., $X_{\mathrm{B}} \geq (L - c_u^{\mathcal{D}}) + \sum_i (\ell_i - m_i^{\mathcal{D}})$.

Proposition 1 (Accounting)**.** *Fix a fan-out and a cache state, and let* $\mathcal{R}{=}\{i : a_i{=}\textsc{Reject}\}$, $\mathcal{H}{=}\{i : a_i{=}\textsc{Hit}\}$, *and* $\mathcal{E}{=}\{i : a_i{=}\textsc{Probe}, e_i{=}1\}$. *Under Assumption 1,*

$$W_{\mathrm{B}} - W = \Delta_{\mathrm{trunk}} + \Delta_{\mathrm{adm}} - \Delta_{\mathrm{ext}}, \qquad (9)$$
$$X_{\mathrm{B}} - X \geq \textstyle\sum_{i\in\mathcal{R}} (\ell_i - m_i^{\mathcal{D}}),$$

with trunk-sharing term $\Delta_{\mathrm{trunk}}{=}W_{\mathrm{B}}^{\mathrm{trunk}} - (L - c_u^{\mathcal{P}})$, *admission term* $\Delta_{\mathrm{adm}}{=}\sum_{i\in\mathcal{R}\cup\mathcal{H}} (\ell_i - m_i^{\mathcal{P}})$, *and probe overhead* $\Delta_{\mathrm{ext}}{=}\sum_{i\in\mathcal{E}} W_i^{\mathrm{ext}}$.

Proof. Sum (5) and (6) over i and add the trunk terms. The baseline charges $\ell_i - m_i^{\mathcal{P}}$ for every i, while PD-Prune charges it only for $i \in \mathcal{A}$ and adds W_i^{ext} for $i \in \mathcal{E}$; the residuals not in $\mathcal{A}$ are exactly $\mathcal{R} \cup \mathcal{H}$. For transfer, HITresiduals satisfy $\ell_i - m_i^{\mathcal{D}}{=}0$, so the baseline lower bound exceeds PD-Prune's transfer by the rejected residuals' decoder-missing pages. □

The decomposition makes explicit which savings depend on the cache regime. Δ_{trunk} is zero in regimes R2–R4 and positive only when concurrent siblings recompute the trunk (R1); Δ_{adm} does not depend on the regime; Δ_{ext} is a cost. Decode-time pruners change only D^+: since they act on generated tokens of residuals that have already been prefilled, W and X do not depend on the pruner. We decompose the stage-local work of a fan-out as

$$C_{\mathrm{fan}} = C_{\mathrm{pre}}(W) + C_{\mathrm{transfer}}(X) + C_{\mathrm{dec}}(D^+) + C_{\mathrm{CoW}} + C_{\mathrm{abort}}, \qquad (10)$$

where C_{CoW} and C_{abort} are page-table operations that alias metadata and do not copy trunk KV. This expression is an accounting abstraction rather than a wall-clock predictor, since execution and transfer may overlap; we therefore measure end-to-end latency directly.

4.4 Pruning-Aware KV State

PD-Prune keeps one copy-on-write context tree per session (Appendix B.2, Figure 2). `fork` aliases the parent page table without copying the trunk KV; only residual pages are newly allocated. Shared pages are reclaimed only after their last live reference is released. A rejected residual allocates nothing; a probe that is not extended frees its residual and probe pages; abort frees the residual pages of every uncommitted candidate, $b\ell_i$ bytes each. On the next observation x, commit binds the child with the longest common prefix,

$$v^\star = \arg\max_v \mathrm{LCP}(x, x_v), \qquad g_u \leftarrow g_u + 1, \qquad (11)$$

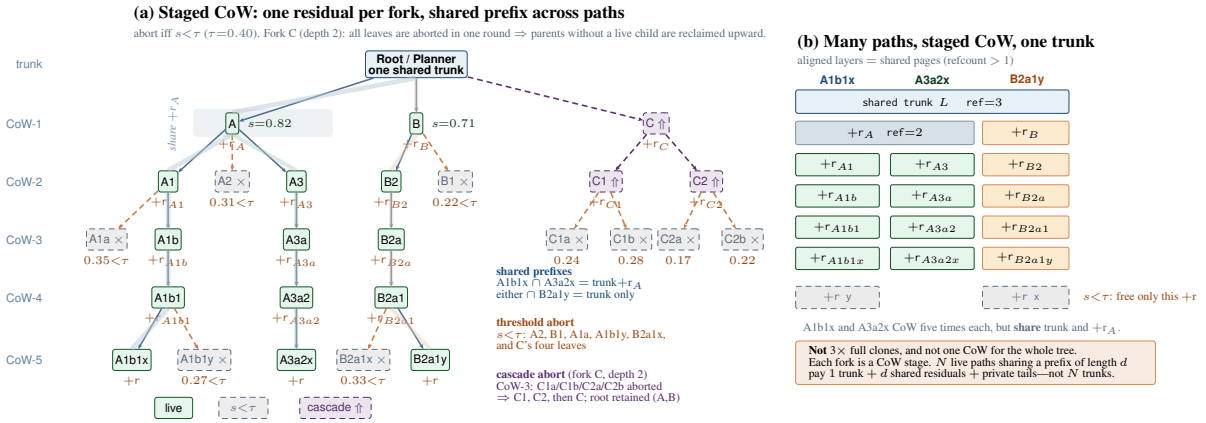


Figure 2: **Context tree.** (a) Each edge is one `fork`: the child aliases the parent and allocates only a residual (+r). Live paths share the trunk; A1b1x and A3a2x also share $+r_A$. All descendants of fork C are aborted at depth 2, so reclamation propagates up to C and stops at the root, which retains the live children A and B. (b) A shared layer is one physical page set with reference count greater than one; aborting a leaf frees only its residual.

publishes the full pages of $x_\star$ to the prefix index, and invalidates in-flight work tagged with an older generation g_u. Speculatively prefilled pages are never sampled from, so the emitted tokens equal those of a baseline that prefills committed tokens only. After commit the live occupancy is M_{spine}, and a pool of M_{pool} bytes holds $\lfloor M_{\text{pool}}/M_{\text{spine}} \rfloor$ sessions instead of $\lfloor M_{\text{pool}}/M_{\text{APC}} \rfloor$ while a prefix cache still retains aborted residuals.

Cross-step coordination. The committed spine of step t becomes the logical trunk of step $t{+}1$. Its KV residency on the prefill and decode instances is restored according to the underlying KV synchronization policy. In our evaluation, the trunk is warm before each fan-out, so the admission results isolate residual-side prefill and transfer savings rather than cross-step trunk synchronization. Once the required trunk pages are resident, subsequent fan-outs can reuse them without recomputing the shared prefix. A residual becomes a HIT only when its parent-aware prefix is already resident on the decode instance. Admission is re-evaluated at every step, so rejected branches never accumulate state across turns.

5 Evaluation

Setup. All GPU runs use vLLM on A100-SXM4 40GB GPUs with `bfloat16` weights, temperature 0, and APC enabled. Prefill and decode are physically disaggregated with a warm trunk (regime R3), so prefill differences reflect residual admission only. Admission and probe results use Qwen3-8B on a *branching mix* of valid, repetitive, and malformed residuals for GSM8K and MATH-500; spine retention uses the full GSM8K test set on Qwen3-14B. We report computed prefill W, decode tokens D^+, peak KV, wall-clock, *committed-answer accuracy*, and *any-finished* coverage. Hyperparameters, mix construction, and run-to-run variation are in Appendix C.

Prefill-Side Admission Complements Decode-Time Pruning. Table 2 (GSM8K block) places one admission policy in front of three decode-time pruners. Without admission, ESC, Speculative Rejection, and DPTS prefill the same 5008 tokens; they differ only in D^+ (131072, 98304, and 78336 tokens) because each reads generated text. Reject-only admission at τ_{pre}=0.15 rejects the four repetitive residuals before any kernel and reduces W to 3848 tokens (-23%) for every pruner, as Proposition 1 predicts. Committed accuracy is unchanged for every pruner and coverage remains $16/16$. The time reduction is largest for ESC (-12%), which would otherwise decode the repetitive residuals to the full budget, and small for pruners that already stop them early (-6%, -3%). Aggressive prefill-side rejection ("Blanket") at 0.45 also removes the malformed residuals and is faster (-34% for ESC); at k=16 coverage stays $16/16$ and committed accuracy drops by at most one problem (DPTS). Table 3 shows that the trade-off depends on width: at k=4 and k=8, Blanket loses one to two problems of coverage, while reject-only stays within one of the pruner-only baseline. At k=32 the same reject-only gate cuts ESC wall-clock by 17% and blanket by 49%, both still at $16/16$ coverage.

Table 2: Main results on Qwen3-8B (branching mix, trunk warm). GSM8K: prefill-side admission composed with decode-side pruners. MATH-500: admission vs. probe under coverage. Settings in Appendix C.

Policy	admitted	W	time (s)	committed	any-finished
GSM8K; columns list ESC / Speculative Rejection / DPTS					
pruner only	16	5008	16.49 / 13.29 / 11.87	9 / 9 / 10	16 / 16 / 16
+ Reject-only	12	3848	14.45 / 12.53 / 11.57	9 / 9 / 10	16 / 16 / 16
+ Blanket	8	2768	10.95 / 10.95 / 10.96	9 / 9 / 9	16 / 16 / 16
MATH-500; out of 32					
full fan-out	16	12320	104.1	3	20
Blanket	8	6200	52.1	3	13
Reject-only	12	9332	71.4	2	16
Probe	8+4	28736	72.2	3	20

Table 3: ESC on GSM8K across fan-out width (Qwen3-8B, n=16). Each cell: committed / any-finished / seconds.

	k=4	k=8	k=16	k=32
ESC	9 / 16 / 8.62	8 / 16 / 10.90	9 / 16 / 16.48	9 / 16 / 32.78
+ Reject-only (0.15)	8 / 16 / 8.22	10 / 16 / 10.25	10 / 16 / 14.45	9 / 16 / 27.14
+ Blanket (0.45)	9 / 14 / 7.90	8 / 15 / 8.66	8 / 16 / 10.95	9 / 16 / 16.61

Table 4: MATH-500 coverage (any-finished out of 32) and seconds across width. "Blanket" is aggressive prefill-side rejection (s_i<0.45); "Reject-only" rejects repetitive residuals; "Probe t_0" probes the band $[0.15, 0.45)$.

	k=4	k=8	k=16
Full fan-out	12, 25.3	15, 52.5	20, 104.1
Blanket	6, 19.1	11, 25.6	13, 52.1
Reject-only	11, 23.0	17, 41.5	16, 71.4
Probe t_0=64	8, 24.0	16, 42.0	18, 72.5
Probe t_0=128	9, 24.7	13, 43.3	20, 72.2

Decode-side probing mitigates the coverage loss of aggressive prefill-side rejection. On MATH-500 (Table 2, lower block), Blanket halves the time of the full fan-out but lowers coverage from $20/32$ to $13/32$: some residuals scored as malformed nevertheless lead to correct solutions once decoded. Reject-only admission reaches $16/32$. At k=16, probing the uncertainty band with t_0=128 increases coverage to $20/32$ at 72.2 s, $0.69\times$ the full fan-out, and reduces D^+ from 524288 to 393216 tokens. Most uncertain residuals pass the decode-side refinement in this workload; Appendix D.5 provides the detailed probe outcomes and recomputation overhead. The cost of probing is visible in W: at k=16, evicted probe pages are recomputed on extension and W rises to 28736 tokens, which (9) accounts for as Δ_{ext}; at $k \leq 8$ the overhead is below 300 tokens. The best probe length depends on width (Table 4): t_0=64 is preferable at k=8, and at k=4 no probe matches the full fan-out.

Table 5: Control-plane connector under P/D disaggregation (not GPU wall-clock). "P/D+rep." withholds repetitive residuals. Detailed rates and session parameters in Appendix C.

	APC	P/D	P/D+rep.	Alias+gate
Prefill W	11520	11520	8640	960
Connector X	0	11520	8640	960
Transfer	0	23.0 ms	17.3 **ms**	1.9 **ms**
Fan-out	138.2 ms	161.3 ms	121.0 **ms**	13.4 **ms**

Connector volume. Table 5 is the control-plane accounting of Proposition 1 for 10 sessions with k=4, L=256, and ℓ_i=32. Stock P/D ships $X{=}kL + \sum_i \ell_i$=11520 tokens. Withholding only the repetitive residuals (τ_{pre}=0.15) on a non-aliased prefiller keeps three residuals per fan-out and cuts both W and X by 25% (8640). Aliasing the trunk on top of that gate removes L from both columns, so the connector carries only the miss tail of $\mathcal{K}$ (960 tokens, $0.08\times$ stock P/D). The same ordering holds at $k \in \{8, 16, 32\}$ (Appendix D.6).

Pruning-aware spine retention lowers post-selection KV occupancy. On the full GSM8K test set (n=1319, Qwen3-14B, tp=2, k=4), retaining only the committed spine matches APC within one point of committed-answer accuracy (0.845 vs. 0.844) and reduces median peak KV usage to $0.72\times$ that of APC (1030 vs. 1430 tokens; Appendix C.1). This result comes from state management alone (CoW-only, i.e., PD-Prune without admission), since admission does not change the committed spine. The ratio follows $(L + \ell_\star)/(L + \sum_i \ell_i)$ and is therefore governed by k and the trunk-to-residual ratio rather than by the model: it is 0.72–$0.75\times$ on GSM8K and 0.57–$0.61\times$ on Game of 24 across six families (Figure 12 in Appendix C.8), and it falls from about 0.90 at k=2 to 0.53–0.55 at k=8 (Appendix C.9). The copy-on-write fan-out incurs its own overhead: its median latency is 140 ms, compared with 87 ms for

Table 6: Three-step Tree-of-Thoughts (n=16, k=4, tp=2). Accuracy is committed-answer accuracy; e2e is seconds per problem; peak is spine tokens. The answer-marker stop in the PD-Prune column is orthogonal and is not attributed to PD-Prune.

	Qwen3-4B			Qwen3-8B		
	APC	CoW-only	PD-Prune	APC	CoW-only	PD-Prune
GSM8K acc.	0.938	0.938	0.938	1.000	0.938	0.938
GSM8K e2e	1.66	1.76	1.63	2.48	2.56	1.91
GSM8K peak	318	275	275	318	275	275
Game24 acc.	0.375	0.500	0.562	0.562	0.562	0.625
Game24 e2e	1.61	1.64	1.13	2.15	2.15	1.52
Game24 peak	265	216	216	265	216	216

an APC hash hit.

Multi-step agent sessions. We run a three-step Tree-of-Thoughts session (k=4 per step, n=16, Qwen3-4B and Qwen3-8B, tp=2) in which each step's spine is the next step's trunk (Table 6). Admission runs before every fan-out and prefills 48 of the 192 residuals. Peak KV is $0.86\times$ APC on GSM8K and $0.82\times$ on Game of 24 for both CoW-only and PD-Prune. Committed accuracy matches APC on Qwen3-4B GSM8K, is one problem lower on Qwen3-8B GSM8K (15 vs. 16 of 16), and is one to three problems higher on Game of 24. The answer-marker stop in the PD-Prune column is an orthogonal optimization and is not attributed to PD-Prune. End-to-end times in Table 6 therefore mix admission with this stop, and we do not treat those reductions as a PD-Prune result.

6 Conclusion

In disaggregated agentic serving, the cost of a discarded continuation is incurred at the prefill instance and the connector, whereas existing pruning signals are evaluated at the decode instance. PD-Prune is a cross-stage pruning framework: it applies pre-kernel admission from residency and a prefill-side heuristic score, defers uncertain candidates to a decode-side probe with a separately calibrated threshold, and uses a pruning-aware copy-on-write context tree to manage the KV state created by these decisions, retaining only the committed spine. The accounting in Proposition 1 separates the savings that depend on the cache regime from those due to admission. Empirically, prefill-side admission is complementary to three decode-time pruners and reduces computed prefill by 23% at matched committed-answer accuracy; at k=16, a 128-token probe increases coverage to $20/32$ at $0.69\times$ the full-fan-out wall-clock; and retaining only the committed spine reduces median peak KV usage to $0.72\times$ that of APC.

Limitations

Our scorers are lightweight heuristics evaluated on a controlled branching mix; natural agent residuals may be less separable, and learned scorers such as STOP [4] can be substituted without changing the admission policy. The admission experiments use moderate sample sizes (n=16 and n=32), so small differences in committed accuracy fall within run-to-run variation. Extending the evaluation from reasoning and Tree-of-Thoughts sessions to tool-using agents with real environments is a natural next step.

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A Background

A.1 Branching in Agentic and Reasoning Workloads

Chain-of-thought produces one trace [32]; Tree-of-Thoughts, self-consistency, and Monte Carlo tree search produce many and then select [35, 31, 39]. In the Game-of-24 setting of Tree-of-Thoughts, a proposal prompt emits candidate equations, a value prompt labels each candidate, and breadth-first search retains a beam [35]. Branching controllers follow the same pattern: a coding or tool-using agent proposes several continuations of one context, executes or observes one, and discards the rest [36, 34, 30]. On GSM8K and MATH the accuracy gain of such width is well documented [8, 12, 19], and so is its token cost [21, 4].

A.2 Decode-Time Path Pruning

Early-stopping self-consistency draws a window of samples and stops once the window agrees [17, 31]. Speculative Rejection ranks partial generations at a reward horizon and halts the lower quantile [26]. DPTS generates a mini-step on every active node and early-stops low-confidence children [10]. STOP organizes these signals by source and learnability and trains an internal token that predicts futility from a short prefix [4]; EquivPruner merges semantically equivalent actions [21]. Each signal is a function of generated text. On our GSM8K mix at k=16, ESC, Speculative Rejection, and DPTS therefore prefill the same 5008 tokens and differ only in decode volume (131072, 98304, and 78336 tokens).

A.3 Prefix Caching and Disaggregated Transfer

APC and RadixAttention index full blocks by chained keys, $h_j = \mathrm{hash}(h_{j-1}, \mathrm{block}_j, \mathrm{extra})$ [14, 41]. A request reuses a block only if the block and its entire parent chain are resident. Whether k siblings issued together share the trunk depends on the cache regime of §3.2: a resident trunk (R3) is shared by all siblings; a cold trunk is computed once if siblings are serialized behind it (R2), and between one and k times if they are admitted concurrently, depending on how the scheduler treats

Table 7: Notation.

Symbol	Domain	Meaning
$L, \ell_i, \ell_\star$	$\mathbb{N}$	trunk; residual i; committed residual
$x_u, \rho_i, x_\star$	tokens	trunk; residual; spine
$\mathcal{P}, \mathcal{D}$	—	prefill and decode instances
$c_u^{\mathcal{X}}, m_i^{\mathcal{X}}$	tokens	resident prefix of trunk and residual on $\mathcal{X}$
s_i, s_i'	$[0, 1]$	residual score; probe score
$\tau_{\text{pre}}, \tau_{\text{dec}}$	$[0, 1]$	rejection and extension thresholds
a_i	$\mathcal{A}$	HIT, ADMIT, PROBE, REJECT(Definition 2)
e_i	$\{0, 1\}$	probe of i is extended
t_0, D, d_i	tokens	probe length; budget; decode tokens of i
W_i, X_i	tokens	computed prefill; connector tokens
W_i^{ext}	tokens	recomputation on probe extension
$\mathcal{R}, \mathcal{H}, \mathcal{E}$	$\subseteq [k]$	rejected, hit, and extended residuals
b, P	bytes, tokens	KV bytes per token; page size
M_{pool}	bytes	KV pool capacity
$B_t = B_t^c + B_t^s$	tokens	committed and speculative chunk budget

blocks that are still being computed (R1). APC thus addresses trunk reuse, but it neither rejects a residual nor decides which pages cross a connector.

PD disaggregation places prefill and decode on separate instances and transfers KV between them [42, 25]. Every admitted request is prefilled on $\mathcal{P}$, and the pages that $\mathcal{D}$ lacks are transferred; a residual that the branching controller later discards has already consumed both. Program- and workflow-level schedulers raise the scheduling unit above the request [23, 13, 16, 33], and KV retention removes trunk recomputation across tool gaps; the remaining per-step cost is the residual suffixes of siblings that the branching controller can enumerate before the observation returns.

A.4 Prefill, Decode, and Idle Time

Prefill processes the prompt in parallel and is compute-bound; decode emits one token per step and is memory-bandwidth-bound [37, 42, 25]. PagedAttention stores KV in non-contiguous pages [14], and chunked prefill co-schedules a prompt chunk with a decode batch [2]. For agents, the prompt of step $t{+}1$ extends that of step t after a tool interval of variable length [16, 33], during which known suffixes can be prefilled (Appendix B.5). Figure 3 shows where each existing mechanism can first discard a sibling, and Table 7 summarizes the notation.

B System Details

PD-Prune sits between the branching controller and the LLM engine (Figure 1). The branching controller exposes branch structure, PD-Prune decides and binds, and the engine executes prefill, transfer, and decode.

B.1 Interface

`open(session)`	Create a root; prefill the initial context.
`fork(u, bid, tok?)`	Child of u, optionally with a known suffix.
`speculate(v, pri)`	Mark v eligible for idle-time prefill.
`commit(u, tokens)`	Bind by LCP ((11)); start decode.
`join(children, pol)`	Merge a fan-out (`all`/`first`/`kofn`).
`abort(v)` / `close`	Drop a subtree; end the session.

Only `commit` lies on the critical path of user-visible tokens.

B.2 Copy-on-Write Context Tree

Definition 3 (Context tree)**.** A session is a rooted tree $T = (V, E)$. Node u stores a token sequence x_u with $x_{\pi(u)} \preceq x_u$, a residual $\delta_u = x_u[|x_{\pi(u)}|{:}]$, a mode in $\{\text{SPEC}, \text{COMMIT}, \text{IDLE}, \text{DEAD}\}$, a generation g_u, and a page table $\sigma_u = (\text{parent}, \text{alias_len}, \text{pages}_u)$.

The tree maintains four invariants: (i) prefix closure; (ii) at most one committed child per node outside a join; (iii) speculative nodes are never sampled from; (iv) a physical page remains live while at least one non-dead node maps it. $\texttt{fork}(u, \cdot)$ creates a metadata alias to the parent page table and registers the child as a live reference; no trunk KV data are copied, and only residual pages are newly allocated. A write to a shared or read-only page copies only its valid rows. Shared pages are reclaimed only after their last live reference is released. A multi-level fan-out is a sequence of such forks (Figure 2); abort frees residual pages only, and a subtree without live descendants is reclaimed up to the nearest ancestor with a live child. Commit publishes the full pages of $x_\star$ to the prefix index so that a later request whose parent-aware prefix is already resident on $\mathcal{D}$ can take HIT. Eviction is node-granular: speculative leaves first, then idle committed leaves, then the spine [16, 33].

B.3 Speculative Prefill of Known Suffixes

A candidate of parent u is a pair $(x^{\text{k}}, \hat{x}^{\text{o}})$ of a known suffix and a predicted observation, with admission probability p and expected prefix-hit rate q. Known suffixes ($p{=}q{=}1$) are scheduled before predicted observations ($q{<}1$), subject to the idle time $T_{\text{idle}}(u)$, the available KV capacity, and a branch cap. Unlike speculative decoding [15, 6], the object prefilled is a future *prompt* prefix rather than draft output tokens. On commit, the pages covered by $\text{LCP}(x, x_{v^\star})$ become committed, the unmatched

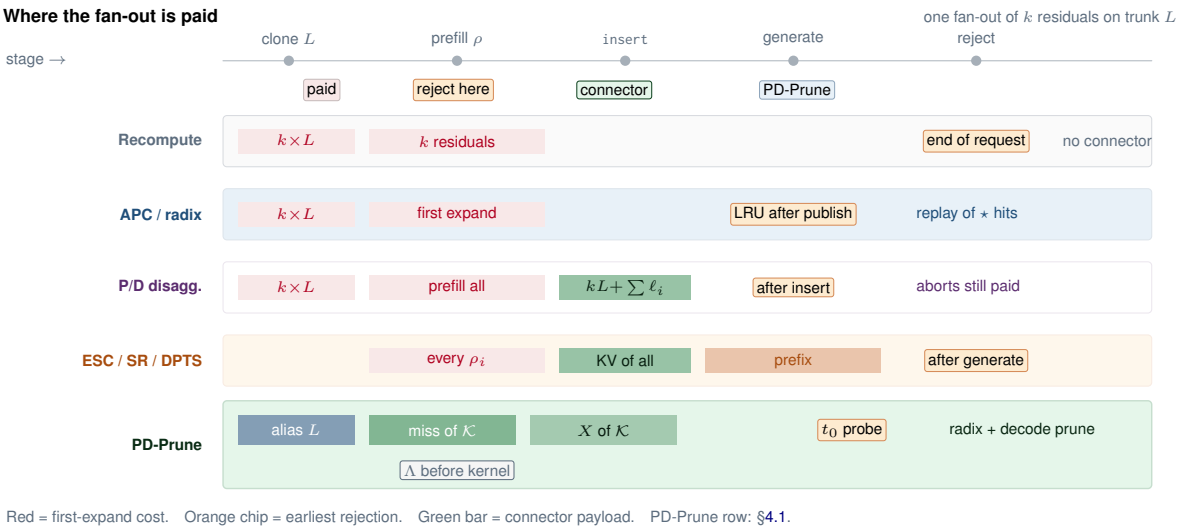


Figure 3: **Where each mechanism acts on one fan-out of k residuals over a trunk of length L.** Red bars denote prefill issued before any sibling is discarded, orange chips the earliest rejection point, and green bars connector volume. Trunk volumes for the prefix-cache rows assume the cold concurrent regime R1; in regimes R2–R4 the trunk is computed at most once (§3.2). The PD-Prune row follows §4.1.

tail is prefilled, every sibling is aborted, and jobs tagged with a stale generation are dropped by generation tag.

B.4 Admission Details

Scorers. $\text{Score}_{\text{pre}}$ assigns 0.02 to a residual containing a substring of 8–40 characters repeated at least four times, 0.22 to a residual containing malformed mathematical tokens (e.g., runs of asterisks or nan), and otherwise a weighted combination of alphanumeric fraction and character diversity, which is close to 1 for natural prose. $\text{Score}_{\text{dec}}$ applies the same repetition and malformation tests to the generated prefix (0.04 and 0.16), assigns 0.92 once a finished answer marker (#### followed by a number, or \boxed{) appears, and otherwise uses a weighted combination that also rewards digits and operators. Both scorers run on the CPU and do not share parameters.

Ordering of the action. (4) reads decode-side residency first, because a residual already resident on $\mathcal{D}$ costs neither prefill nor transfer. It then protects index 0, so the top-ranked continuation is always served. Only then does it consult the score. A page resident only on $\mathcal{P}$ reduces W_i but not X_i; the residual is admitted with $W_i{=}\ell_i - m_i^{\mathcal{P}}$ and $X_i{=}\ell_i - m_i^{\mathcal{D}}$. Figure 5 illustrates the four actions on one radix tree, and Figure 6 places them on the request path.

Spans. Table 8 summarizes prefill, transfer, and post-abort memory per fan-out for a cold trunk.

B.5 Critical-Path Placement

After abort the live occupancy is M_{spine}, so a pool of M_{pool} bytes holds

$$C = \lfloor M_{\text{pool}}/M \rfloor \tag{12}$$

sessions, with $M{=}M_{\text{spine}}$ for PD-Prune and $M{=}M_{\text{APC}}$ while a prefix cache retains aborted residuals. On tick t the chunk budget [2] is split by the committed rate $\underline{s}_r$ [40],

$$B_t^{\text{c}} = \min\Big(B_t,\ \sum_{r \in R_t^{\text{c}}} \max(\underline{s}_r \Delta,\ 1)\Big), \tag{13}$$

$$B_t^{\text{s}} = B_t - B_t^{\text{c}}. \tag{14}$$

B_t^{c} decodes committed spines; B_t^{s} prefills admitted residuals or known suffixes ($p{=}1$) when $T_{\text{pre}}(|w|) \le T_{\text{idle}}(u)$. Speculatively prefilled tokens are never sampled from.

B.6 Multi-Step Pipeline

Algorithm 2 shows how Alg. 1 repeats across agent steps on the PD path: admit on the controller, prefill on $\mathcal{P}$, ship on the connector, then probe and decode on $\mathcal{D}$. The committed spine of step t becomes the logical trunk of step $t{+}1$. Its KV residency on $\mathcal{P}$ and $\mathcal{D}$ is restored according to the underlying KV synchronization policy; in our evaluation the trunk is warm before each fan-out.

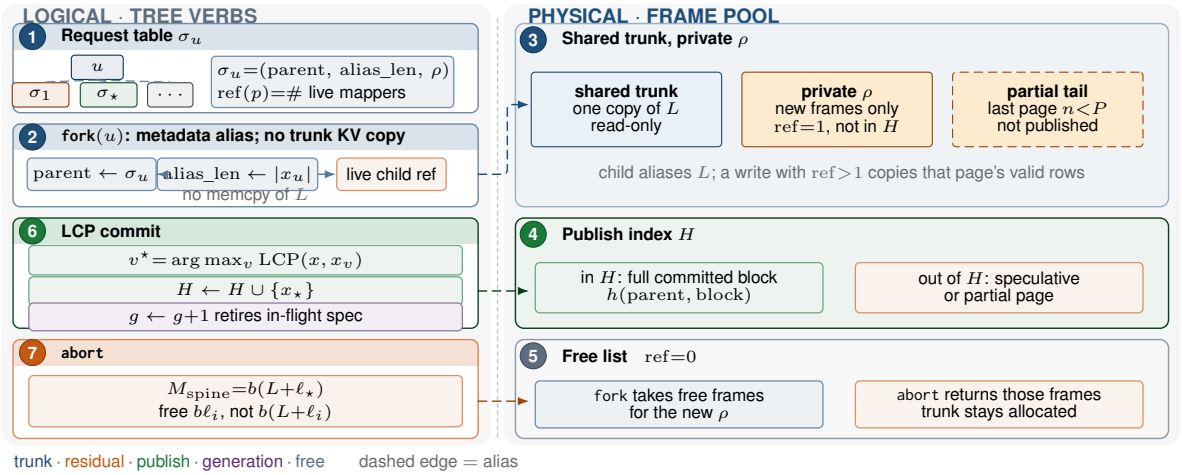


Figure 4: **Context-tree page pool** (pruning-aware KV state). `fork` aliases the parent page table without copying the trunk KV and allocates only residual pages ①–③. LCP commit publishes full blocks of $x_\star$ into H so a later request whose parent-aware prefix is already resident on $\mathcal{D}$ may take HIT ④⑥. `abort` returns residual pages only; the live occupancy is $M_{\text{spine}}=b(L+\ell_\star)$ ⑤⑦.

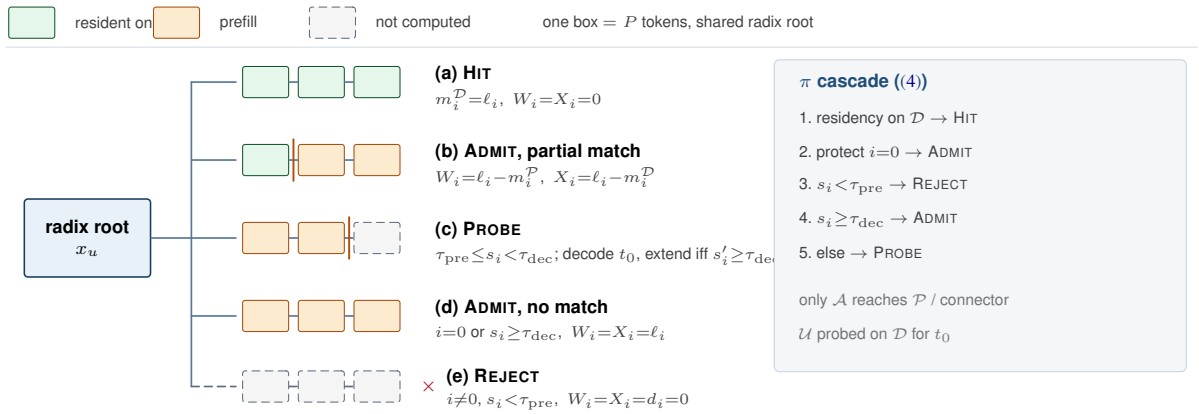


Figure 5: **Policy π on one radix tree** ((4), Alg. 1). The root is the aliased trunk x_u; each box is one page of P tokens. Green pages are already resident on $\mathcal{D}$, orange pages are the miss prefilled on $\mathcal{P}$, and gray pages are never computed. The cascade on the right is the controller decision order: residency, protect $i=0$, then $(\tau_{\text{pre}}, \tau_{\text{dec}})$. Only $\mathcal{A}$ reaches $\mathcal{P}$ and the connector; $\mathcal{U}$ is probed on $\mathcal{D}$ for t_0 tokens.

C Additional Evaluation

This appendix reports the full experimental setup, the GSM8K forest on Qwen3-14B (§C.1), details of the admission and probe experiments (§C.2–§C.5), a Qwen3-8B forest with an answer-marker decode stop (§C.6), the multi-step sessions (§C.7), cross-model results (§C.8), and sensitivity to width, depth, and trace length (§C.9).

Common setup. All GPU runs use vLLM on A100-SXM4 40GB GPUs with `bfloat16` weights, temperature 0, and APC enabled. Prefill and decode are physically disaggregated: $\mathcal{P}$ and $\mathcal{D}$ run as separate vLLM instances connected by a KV connector, and admitted residuals reach $\mathcal{D}$ only through that connector. We measure computed prefill W, decode tokens D^+, peak KV, and end-to-end wall-clock. The trunk is warmed before every fan-out (regime R3), so prefill differences reflect residual admission only. Models include Qwen3-14B, Qwen3-8B, Qwen3-4B, Qwen2.5-Math-1.5B, Llama-3-8B-Instruct, Mistral-7B-Instruct, and DeepSeek-R1-Distill-Llama-8B (R1-8B).

Admission and probe workloads. The admission experiments use Qwen3-8B at tp=1 on a *branching mix*: for each problem, half of the k residuals are distinct solution strategies (valid), a quarter are repetitive (repetition loops), and a quarter are syntactically malformed. GSM8K uses n=16 problems and D=512; MATH-500 uses

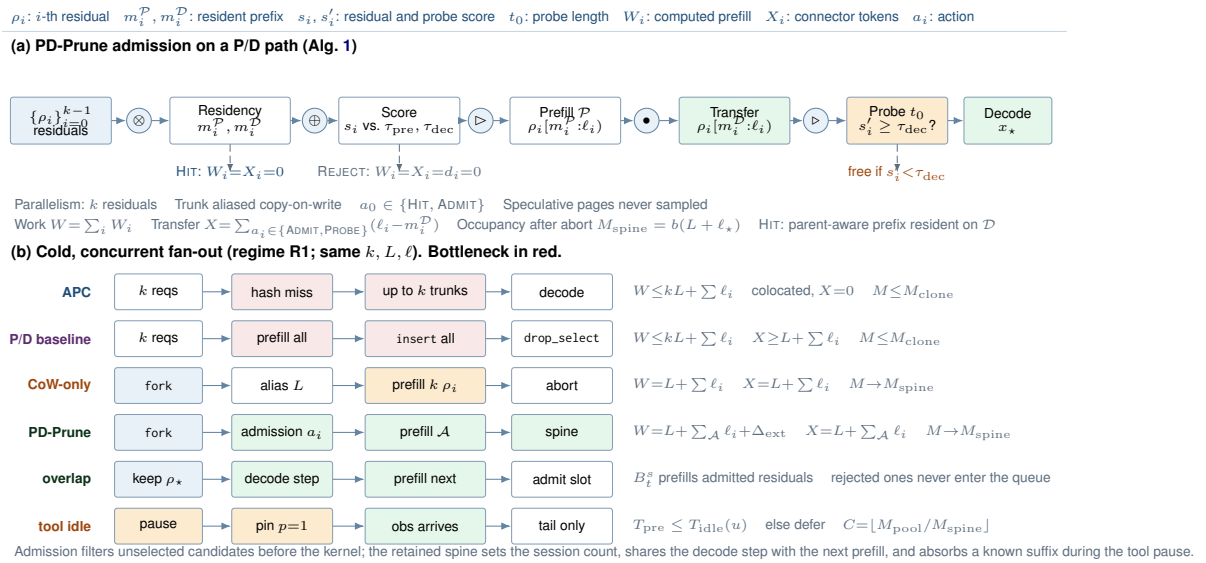


Figure 6: **Admission on the request path.** (a) The action of (4) is evaluated before `insert`; only admitted and probed residuals reach the prefill kernel. (b) Prefill and connector volume of a cold, concurrent fan-out (regime R1) for APC, a baseline PD deployment, the copy-on-write tree without admission (CoW-only), and PD-Prune. The overlap row prefills the next residual during decode; the tool-interval row prefills a known suffix while $T_{\text{pre}} \leq T_{\text{idle}}$.

Table 8: Per-fan-out spans with a cold trunk ($c_u^{\mathcal{P}}{=}c_u^{\mathcal{D}}{=}0$) and no resident residual pages. Baseline rows assume concurrent admission without deduplication (R1); with serialized admission (R2) their trunk term becomes L. $\mathcal{A}$ is the set of admitted and probed residuals; T_{ab} is the abort cost in pages; "replay" is a later request for the committed spine.

	W	X	M after abort	T_{ab}	trunk copies	replay W
Recompute	$kL+\sum_i \ell_i$	0	$kL+\sum_i \ell_i$	$\Theta(kL/P)$	k	$L+\ell_\star$
APC (R1)	$\leq kL+\sum_i \ell_i$	0	$\leq kL+\sum_i \ell_i$	LRU	1–k	0
PD baseline (R1)	$\leq kL+\sum_i \ell_i$	$\geq L+\sum_i \ell_i$	$\leq kL+\sum_i \ell_i$	LRU	1–k	0
CoW-only	$L+\sum_i \ell_i$	$L+\sum_i \ell_i$	$L+\ell_\star$	$\Theta(\ell_i/P)$	1	0
PD-Prune	$L+\sum_{i\in\mathcal{A}} \ell_i+\Delta_{\text{ext}}$	$L+\sum_{i\in\mathcal{A}} \ell_i$	$L+\ell_\star$	$\Theta(\ell_i/P)$	1	0

Algorithm 2 PD-Prune multi-step pipeline. Each step invokes Alg. 1.

Require: initial trunk x_u; thresholds $(\tau_{\text{pre}}, \tau_{\text{dec}}, t_0, D)$
Ensure: committed spines $\{x_\star^{(t)}\}$
OpenTree(x_u)
for each agent step t **do**
  $\{\rho_i\} \leftarrow \text{Expand}(x_u; k)$ ▷ branching-controller fan-out
  Controller: run Alg. 1 $\to x_\star^{(t)}$, (W, X, D^+)
  $\mathcal{P}$**:** prefill only $\mathcal{P}$-miss pages of admitted / probed residuals
  Connector: ship only $\mathcal{D}$-miss pages
  $\mathcal{D}$**:** probe $\to$ decode-time prune $\to$ decode survivors
  $x_u \leftarrow x_\star^{(t)}$ ▷ spine becomes next logical trunk
end for
return $\{x_\star^{(t)}\}$

the first 32 problems of level ≥ 4 and $D{=}1024$. On GSM8K, "Reject-only" rejects $s_i{<}0.15$ and admits the rest; "Blanket" rejects $s_i{<}0.45$. On MATH-500 (no decode-time pruner), "Probe" uses $(\tau_{\text{pre}}, \tau_{\text{dec}}, t_0) = (0.15, 0.45, 128)$, and its W includes W^{ext} from evicted probe pages; the committed residual is fixed to index 0, so committed accuracy is insensitive to admission by construction and coverage is the informative metric. Prefill W is identical across decode-time pruners because those pruners act after prefill.

Metrics. Our primary metric is *committed-answer accuracy*: the answer of the candidate the branching controller commits (index 0 for MATH-500, the pruner's selection for GSM8K) matches the reference. We also report *any-finished* coverage, i.e., whether some full-budget branch is correct, which upper-bounds what a better selector could commit. An answer is correct when its final number matches the reference. With $n{=}16$, identical ESC configurations at $k{=}16$ commit between $8/16$ and $11/16$ across runs (Appendix D.2), so differences of up to three problems are within run-to-run variation. Tree-of-Thoughts forests elsewhere use $k{=}4$ and $D{=}512$ (2048 for contest problems) in batches of eight.

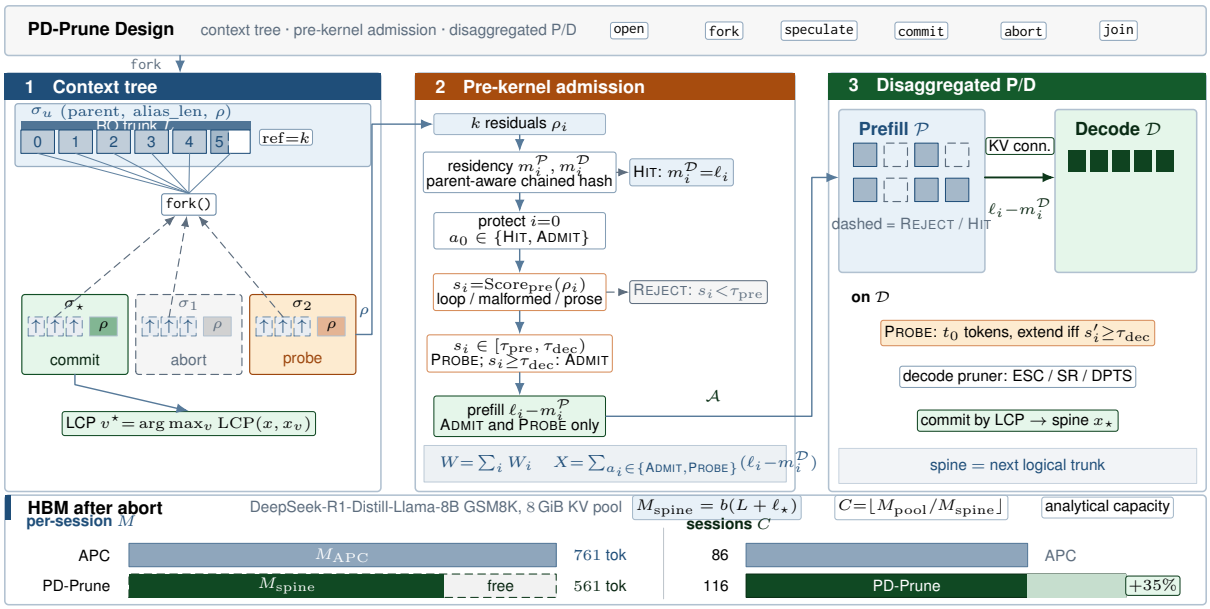


Figure 7: **PD-Prune design.** (1) CoW state: σ_u aliases one trunk, `fork` copies no trunk KV, and LCP commit publishes $x_\star$. (2) Controller policy π: observation $o_i = (m_i^{\mathcal{P}}, m_i^{\mathcal{D}}, s_i)$ maps to $a_i \in \{\text{HIT}, \text{ADMIT}, \text{PROBE}, \text{REJECT}\}$; only $\mathcal{A}$ enters the GPU path. (3) Disaggregated execution: $\mathcal{P}$ prefills the miss of $\mathcal{A}$, the connector ships decoder-missing pages, and $\mathcal{D}$ runs the t_0 probe then the decode-time pruner before LCP commit. Bottom: M_{spine} frees HBM and raises analytical session capacity.

Table 9: GSM8K test, Qwen3-14B, tp=2, n=1319; medians over 165 batches of eight.

	recompute	APC	CoW-only
Committed accuracy	0.847	0.844	0.845
Peak KV, median (tok)	4136	1430	1030
Peak KV, max (tok)	4944	1632	1232
Peak / APC	2.89×	1.00×	0.72×
Fan-out, median (ms)	326	87	140

Connector microbenchmark. Table 5 uses a control-plane accounting over 10 sessions with k=4, L=256, and ℓ_i=32, charging prefill at 12 μs/token and transfer at 2 μs/token. "P/D+rep." is τ_{pre}=0.15 on a stock (non-aliased) prefiller and withholds repetitive residuals; "Alias+gate" aliases the trunk and ships only the miss of $\mathcal{K}$. These numbers are not GPU wall-clock.

C.1 GSM8K Forest on Qwen3-14B

The full GSM8K test set (n=1319) is run on Qwen3-14B at tp=2 in 165 batches of eight (Table 9). This experiment isolates state management: no residual is rejected, so the copy-on-write tree is compared with recomputation and APC on the same committed path.

Median peak KV is 0.72× APC and 0.25× recomputation, and accuracy is within 0.3 points of both. Fan-out latency does not follow the same order: the copy-on-write fan-out takes 140 ms against 87 ms for APC, because fork and abort run on the expand path while an APC hit does not. The same peak ordering (0.58–0.80× APC) holds on seven math datasets for Qwen3-14B, Qwen3-8B, and R1-8B at tp=2 and 4, and peak KV does not change with tensor-parallel width, consistent with (1).

C.2 Probe Admission on MATH-500

The MATH-500 experiments use Qwen3-8B at tp=1, the first 32 problems of level 4 or 5, and D=1024. Half of the k residuals are contest strategies; the other half alternate repetitive and malformed text. Coverage counts a problem when some branch that ran all 1024 tokens is correct; the committed residual is index 0. Aggressive prefill-side rejection ("Blanket") lowers coverage from 12/32 to 6/32 at k=4 and from 20/32 to 13/32 at k=16, whereas no repetitive residual yields a correct answer at any k, so rejecting repetitive residuals does not reduce coverage. At k=16 a 128-token probe increases coverage to 20/32 in 72.2 s instead of 104.1 s. Most uncertain residuals pass the decode-side refinement in this mix; probing the repetitive residuals as well extends none of them and costs 74.4 s. At k=8, t_0=64 reaches 16/32 in 42.0 s, exceeding the full fan-out (15/32 in 52.5 s); at k=4 the t_0=64 probe reaches 8/32 in 24.0 s, and reject-only admission (11/32 in 23.0 s) is the

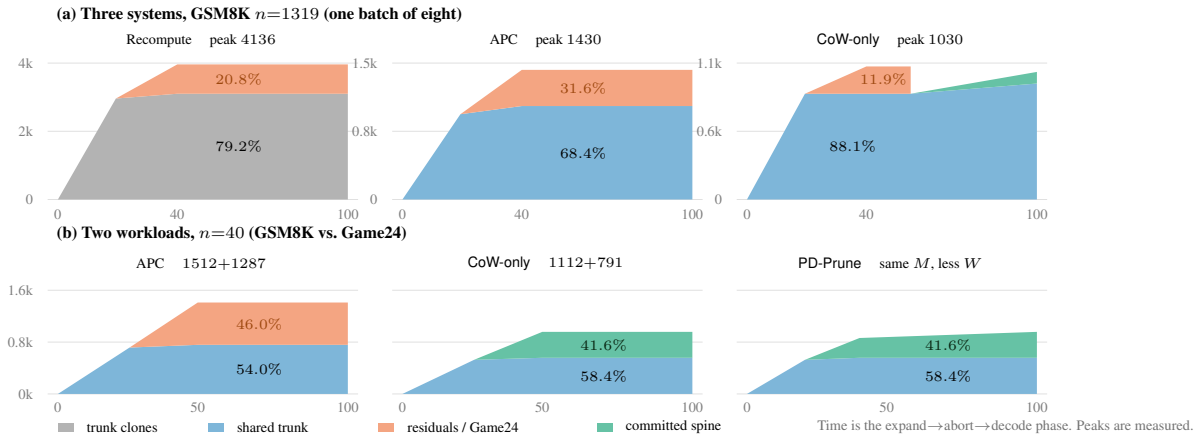


Figure 8: **Stacked KV occupancy** over expand, abort, and decode. (a) One GSM8K batch of eight on Qwen3-14B (per-panel y-scale; peaks 4136, 1430, 1030). (b) GSM8K (bottom) and Game of 24 (top), n=40, for APC, CoW-only, and PD-Prune; CoW-only and PD-Prune share M, and PD-Prune computes fewer residual tokens.

closest non-probe setting. Figure 9 plots Table 4.

C.3 Admission on GSM8K

The GSM8K admission experiments use the branching mix with n=16 and D=512 on Qwen3-8B at tp=1. Because the heuristic scores take three values on this mix (repetitive 0.02, malformed 0.22, valid $\approx$ 1), any $\tau_{\text{pre}} \in (0.02, 0.22]$ yields the same admission set as 0.15, and any $\tau_{\text{pre}} \in (0.22, 1)$ yields the same set as 0.45; a denser threshold sweep therefore adds no information on this workload. Figure 10 plots ESC across widths and the three pruners at k=16. At k=8, reject-only admission keeps ESC at 16/16 coverage (10.84 to 10.19 s) and raises Speculative Rejection and DPTS from 15/16 to 16/16.

C.4 Decode-Only, Prefill-Only, and Combined Pruning

Table 16 (Appendix D.3) factorizes the k=16 results. Decode-only pruning leaves W at 5008 tokens; prefill-only admission decodes every admitted residual to D; the combination retains the prefill reduction and adds the pruner's decode-time termination. At τ_{pre}=0.45 the pruners terminate no additional residuals, because every residual they would terminate has already been rejected.

C.5 Score-Based versus Top-1-Only Admission

Admitting only index 0 is a different policy from scoring every residual. At k=16 and τ=0.45, Top-1-only admission prefills 414 tokens but lowers coverage to 11/16 under ESC and 9/16 under Speculative Rejection, while score-based admission of the eight residuals above τ prefills 2768 tokens and retains 16/16 (Table 17). Committed accuracy is similar for both policies (9–11 of 16), so the difference lies in the coverage available to a better selector.

Capacity from measured peaks. For R1-8B on GSM8K (n=16, k=4), measured median peaks are 761 tokens under APC and 561 under the context tree. With 32 layers, 8 KV heads of dimension 128, and `bfloat16`, one token occupies 131072 bytes, so an 8 GiB KV budget holds 65536 tokens, i.e., 86 APC trees or 116 spines by (12) (+35%). This is an analytical capacity bound; we do not report SLO-constrained throughput, which would additionally depend on scheduling and batching.

C.6 Qwen3-8B Forest with an Answer-Marker Stop

This experiment runs Qwen3-8B on GSM8K with n=80 and D=512 at tp=2 and 4; Game of 24 uses Tree-of-Thoughts with k=4, and HumanEval uses a ReAct wrap-and-recover loop with k=2. The PD-Prune runs additionally stop committed decode when the last non-empty line contains a finished answer, which reduces D^+ to about one third. The answer-marker stop is an orthogonal optimization and is not attributed to PD-Prune. On this trace the scorer admits every issued residual, so the end-to-end ratio in Table 10 reflects the decode stop rather than admission.

Peak KV is 0.75× APC on GSM8K, 0.62× on Game of 24, and 0.96× on HumanEval, where k=2 makes the residual a small fraction of the context. Accuracy is 1–3 points below APC (0.863

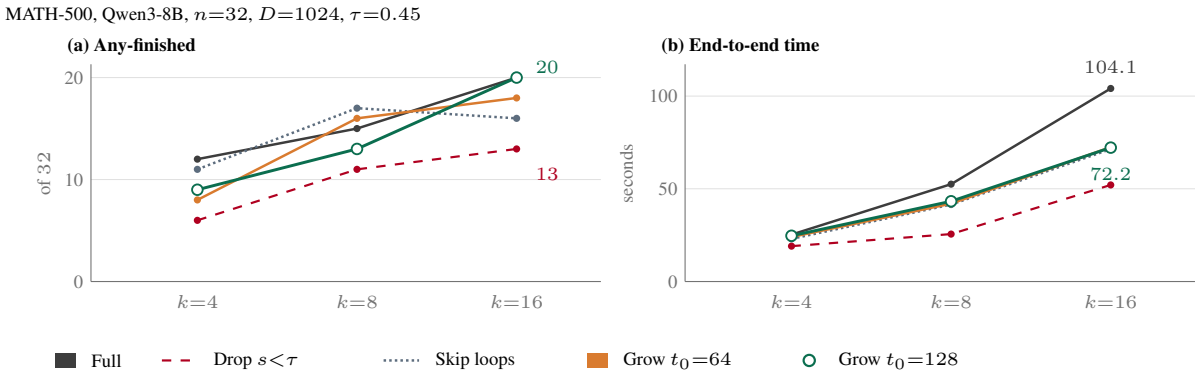


Figure 9: **Probe admission on MATH-500** (Table 4). Blanket rejection is fast but loses coverage; a t_0=128 probe increases coverage to 20/32 at k=16 in 72.2 s instead of 104.1 s.

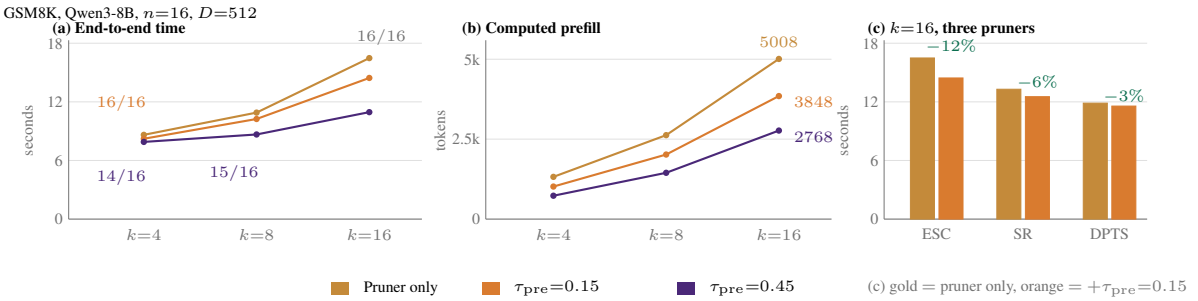


Figure 10: **Admission threshold on GSM8K** (Tables 2 and 3). (a) ESC wall-clock across k, labeled with any-finished coverage. (b) Computed prefill: −23% at τ_{pre}=0.15 and −45% at 0.45. (c) Reject-only admission in front of each pruner at k=16.

Table 10: Qwen3-8B forest, n=80, D=512. PD-Prune includes an answer-marker decode stop. Peak KV does not depend on tp; G24 and HE are the Game of 24 and HumanEval peaks.

	peak	D^+	e2e (s)	acc.	e2e/APC	G24	HE
APC, tp=2	1582	40960	36.0	0.900	—	1289	1970
PD-Prune, tp=2	1182	13356	28.2	0.863	0.78	793	1882
APC, tp=4	1582	40960	22.7	0.938	—	1289	1970
PD-Prune, tp=4	1182	12826	16.9	0.925	0.75	793	1882

vs. 0.900 at tp=2; 0.925 vs. 0.938 at tp=4), which we attribute to the decode stop truncating a few answers.

C.7 Multi-Step Tree-of-Thoughts

Each problem is one session of k=4 candidates over three steps on Qwen3-4B and Qwen3-8B (tp=2, n=16), issued as its own request. Steps 1 and 2 decode a 32-token step, abort the non-selected candidates, and promote the selected spine to the next trunk; step 3 decodes the final answer with D=512 (GSM8K) or D=256 (Game of 24). On Game of 24 the steps propose an equation, continue on the remaining cards, and write the final expression; on GSM8K they set up the algebra, revise the previous step, and state the final number. Admission runs before every fan-out, and the 32-token step is decoded only for admitted residuals. Per-phase timings are not comparable across systems, because the APC baseline attributes step decoding to the expansion phase whereas ours attributes it to decode; Table 6 therefore reports end-to-end time only.

C.8 Cross-Model Results

The forest is repeated on six families (n=40, D=512, tp=2 and 4). APC retains the shared trunk and all residuals, $M_{\text{CoW}} = b(L + \sum_i \ell_i)$, while the context tree retains $M_{\text{spine}} = b(L + \ell_\star)$; their ratio $(L + \ell_\star)/(L + \sum_i \ell_i)$ is independent of the model and of tensor-parallel width. For similar residual lengths it reduces to $(L + \ell)/(L + k\ell)$, which decreases as L/ℓ decreases: 0.72–0.75 on GSM8K with a long trunk and 0.57–0.61 on Game of 24 with a short one (Figure 12). Relative to recomputation, which stores M_{clone}, the spine is 0.25×.

Figure 13 repeats the comparison at n=80; Figure 14 shows per-item decode lengths.

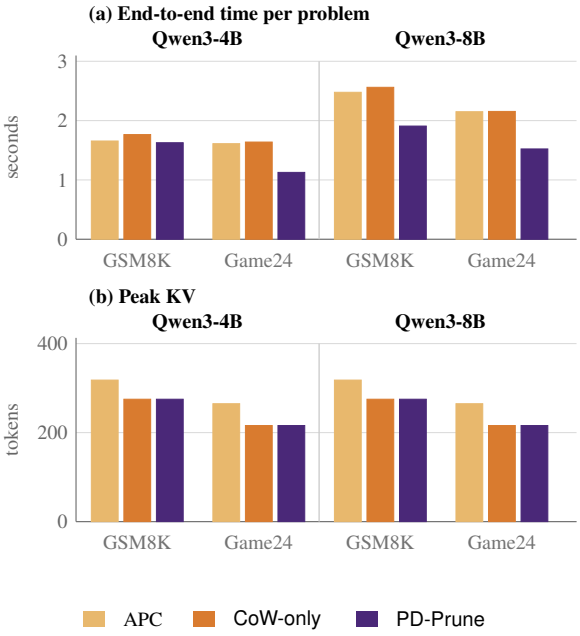


Figure 11: **Three-step Tree-of-Thoughts** (n=16, tp=2). (a) End-to-end seconds per problem; the answer-marker stop in PD-Prune is orthogonal and is not attributed to PD-Prune. (b) Peak KV of the committed spine.

Table 11 isolates the answer-marker stop on three further models (n=32). The stop reduces D^+ by 23–69% and leaves the spine peak unchanged, but its effect on accuracy is model-dependent: +2 problems on R1-8B, −1 on Llama-3-8B, and −5 on Mistral-7B (0.344 to 0.188), where the model emits intermediate answer markers before finishing. The stop is therefore not a safe default, is not part of the admission results in §5, and is not attributed to PD-Prune.

C.9 Width, Depth, and Long Traces

GSM8K runs use n=16, D=512, and tp=2 on Llama-3-8B, Mistral-7B, and R1-8B, varying k or adding a second Tree-of-Thoughts level (Table 12 and Figure 15). The spine peak is invariant in k, while the APC peak grows with the union of residuals, so the ratio falls from 0.89–0.90 at k=2 to 0.53–0.55 at k=8; at depth 2 it is 0.75–0.76. Admission leaves the peak unchanged; at k=8 it rejects 28 branches and reduces fan-out time relative to CoW-only from 307 to 179 ms on Llama-3-8B (APC: 224 ms), 349 to 191 ms on Mistral-7B, and 363 to 194 ms on R1-8B. With a batch of 16, the ratio is 0.72–0.73.

Table 13 reports contest problems with D=2048 (MATH-500 n=200, AIME n=73). The peak ratio is 0.78–0.91, and accuracy is within 3 points of APC on MATH-500 and within one problem on AIME. We do not compare end-to-end time here: the APC baseline decodes the full budget for every item, whereas ours stops at EOS, so time differences would reflect the drivers rather than the methods. The trunk token counts of the two setups also differ slightly (e.g., 95 vs. 73 on R1-8B MATH-500) because of prompt templating, which adds an offset to the ratios.

Table 11: Spine peak and an optional answer-marker decode stop (n=32, tp=2; GSM8K D=512, Game of 24 D=256). The PD-Prune column applies both the context tree and the stop.

Model	Metric	APC	PD-Prune
R1-8B	GSM8K peak	1493	1093
	GSM8K acc.	0.500	0.562
	GSM8K D^+	16384	12693
	Game24 peak	1064	616
Llama-3-8B	GSM8K peak	1485	1085
	GSM8K acc.	0.719	0.688
	GSM8K D^+	16384	5101
	Game24 peak	1224	736
Mistral-7B	GSM8K peak	1521	1121
	GSM8K acc.	0.344	0.188
	GSM8K D^+	16384	6723
	Game24 peak	1296	728

Table 12: GSM8K peak KV (tokens) versus branching (n=16, tp=2, D=512). PD-Prune has the same peak as CoW-only.

Model		k=2	k=4	k=8	depth 2
Llama-3-8B	APC	610	750	998	814
	CoW-only	550	550	550	614
	ratio	0.90	0.73	0.55	0.75
Mistral-7B	APC	629	769	1033	841
	CoW-only	569	569	569	641
	ratio	0.90	0.74	0.55	0.76
R1-8B	APC	629	761	1053	841
	CoW-only	561	561	561	641
	ratio	0.89	0.74	0.53	0.76

D Full Ablation Tables

This appendix gives the full ablations behind §5. All runs use Qwen3-8B (tp=1, temperature 0, APC) on the branching mix of §5: GSM8K with n=16, D=512, and MATH-500 (first 32 problems of level $\geq$ 4) with D=1024. We report computed prefill W (prompt minus cache hits), decode tokens D^+, wall-clock T_{e2e} in seconds (trunk warm-up plus fan-out), committed accuracy (comm.), and any-finished coverage (any).

Notation for this appendix.

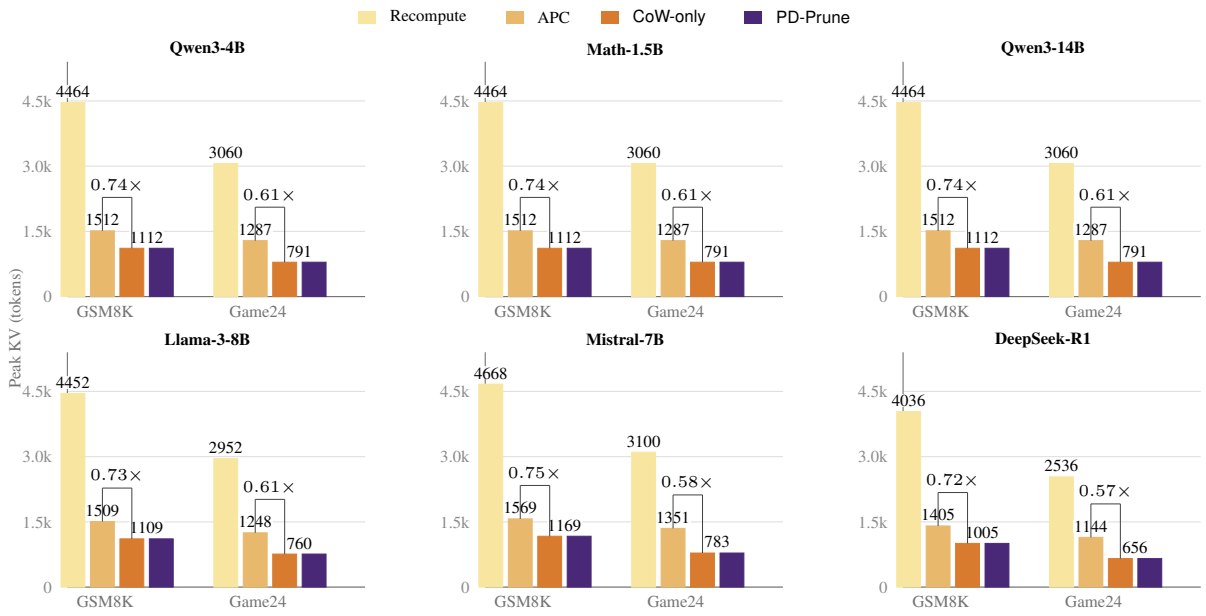


Figure 12: **Peak KV across six model families** (n=40, tp=2). Bars are recompute, APC, CoW-only, and PD-Prune; CoW-only and PD-Prune retain the same spine. The annotated ratio is $M_{\text{spine}}/M_{\text{APC}}$.

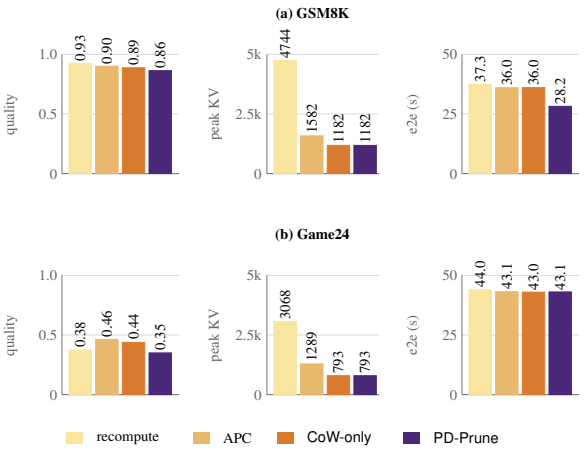


Figure 13: **Four systems on Qwen3-8B** (n=80, tp=2, D=512). Each panel has its own axis; for peak KV and end-to-end time shorter is better. (a) GSM8K. (b) Game of 24. CoW-only and PD-Prune lower peak KV to 0.75× (GSM8K) and 0.62× (Game of 24) APC; the end-to-end reduction of PD-Prune on GSM8K (36.0 to 28.2 s) is due to the answer-marker stop, which is orthogonal and is not attributed to PD-Prune.

Symbol	Domain	Meaning
k	$\{4, 8, 16, 32\}$	fan-out width
τ_{pre}	$\{0, 0.15, 0.45\}$	rejection threshold on $\text{Score}_{\text{pre}}(\rho_i)$
τ_{dec}	0.45	extension threshold on $\text{Score}_{\text{dec}}(y_i)$
t_0	$\{64, 128\}$	probe length (MATH-500)
admit	$\mathbb{N}$	residuals that are admitted or probed
W	tokens	computed prefill
D^+	tokens	decode tokens issued
T_{e2e}	s	end-to-end wall-clock
pruner	$\{\text{ESC}, \text{SR}, \text{DPTS}\}$	decode-time pruner

Table 13: Contest problems, D=2048, k=4, tp=2. All rows of a model come from the same run.

Setting	System	peak	/ APC	acc.
R1-8B MATH	APC	1863	—	0.560
	CoW-only	1599	0.86	0.530
	PD-Prune	1599	0.86	0.530
R1-8B AIME	APC	2118	—	0.041
	CoW-only	1662	0.78	0.041
	PD-Prune	1662	0.78	0.041
Qwen3-4B MATH	APC	1937	—	0.620
	CoW-only	1770	0.91	0.625
	PD-Prune	1770	0.91	0.600
Qwen3-4B AIME	APC	2150	—	0.027
	CoW-only	1794	0.83	0.041
	PD-Prune	1794	0.83	0.041

D.1 Pruner × τ_{pre} × k (GSM8K)

Table 14 is the complete pruner × threshold × width matrix. At $k \leq 16$ the three base pruners report identical W, since pruners act after prefill (§4.3); at k=32 measured W diverges slightly across pruners (10528/10176/9856). Rejecting repetitive residuals (τ_{pre}=0.15) removes a quarter of the residuals and keeps coverage within one problem of the pruner-only baseline. Blanket, aggressive prefill-side rejection at 0.45, removes half of the residuals and is faster, but lowers coverage at k=4 and k=8 for ESC.

Analysis. At $k \leq 16$, the three base pruners report identical W and admit counts. Admission changes only that column: $3k/4$ residuals are admitted at τ_{pre}=0.15 (valid and malformed; repeti-

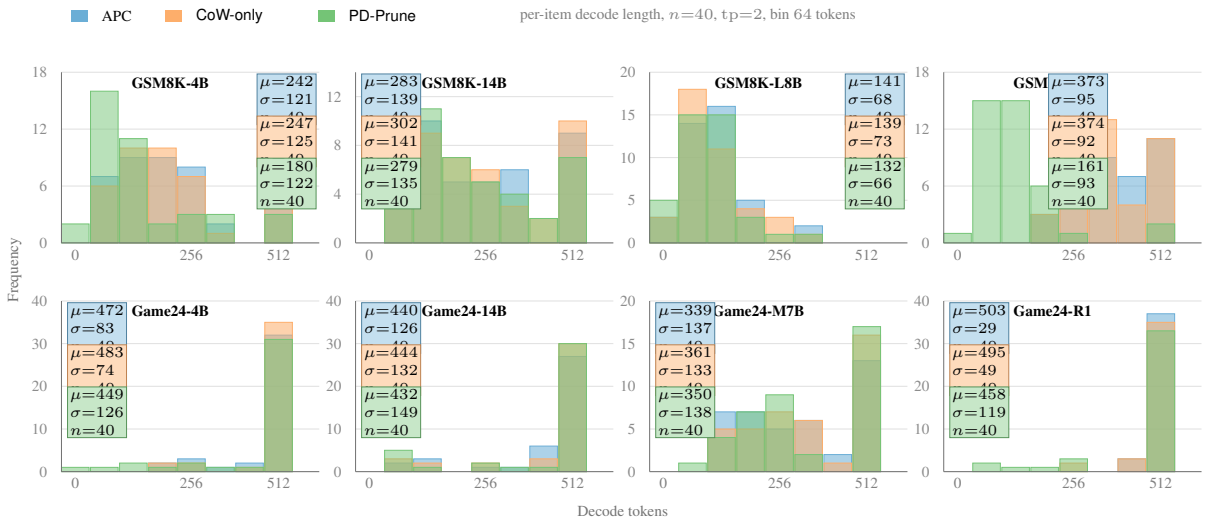


Figure 14: **Per-item decode-length histograms** (n=40, tp=2, D=512, bin width 64). GSM8K (top) and Game of 24 (bottom) overlay APC, CoW-only, and PD-Prune; boxes report μ, σ, and n. Llama-3-8B on Game of 24 is omitted because every output has length 1; Mistral-7B is shown instead.

Table 14: Full decode-time pruner ablation on GSM8K (Qwen3-8B, branching mix, n=16, D=512). Each cell: any / comm. / T_{e2e} s / W / admit. Base rows have no admission; other rows reject $s_i < \tau_{\text{pre}}$ and admit the rest.

Pruner	Gate	k=4	k=8	k=16	k=32
ESC	base	16/16, 8/16, 8.63, 1320, 4	16/16, 8/16, 10.91, 2624, 8	16/16, 9/16, 16.49, 5008, 16	16/16, 9/16, 32.80, 10528, 32
ESC	τ_{pre}=0.15	15/16, 9/16, 8.23, 1018, 3	16/16, 10/16, 10.26, 2020, 6	16/16, 9/16, 14.45, 3848, 12	16/16, 9/16, 27.15, 7968, 24
ESC	τ_{pre}=0.45	14/16, 9/16, 7.91, 732, 2	15/16, 8/16, 8.67, 1448, 4	16/16, 9/16, 10.95, 2768, 8	16/16, 9/16, 16.60, 5408, 16
SR	base	13/16, 9/16, 8.21, 1320, 4	15/16, 9/16, 9.64, 2624, 8	16/16, 9/16, 13.29, 5008, 16	16/16, 10/16, 26.89, 10176, 32
SR	τ_{pre}=0.15	12/16, 9/16, 8.05, 1018, 3	15/16, 9/16, 9.39, 2020, 6	16/16, 9/16, 12.53, 3848, 12	16/16, 9/16, 24.81, 7744, 24
SR	τ_{pre}=0.45	14/16, 9/16, 7.91, 732, 2	15/16, 8/16, 8.69, 1448, 4	16/16, 9/16, 10.95, 2768, 8	16/16, 9/16, 16.59, 5408, 16
DPTS	base	13/16, 10/16, 8.03, 1320, 4	15/16, 10/16, 9.06, 2624, 8	16/16, 10/16, 11.87, 5008, 16	16/16, 9/16, 23.20, 9856, 32
DPTS	τ_{pre}=0.15	13/16, 7/16, 7.98, 1018, 3	15/16, 10/16, 8.96, 2020, 6	16/16, 10/16, 11.57, 3848, 12	16/16, 10/16, 21.01, 7568, 24
DPTS	τ_{pre}=0.45	14/16, 9/16, 7.92, 732, 2	15/16, 8/16, 8.67, 1448, 4	16/16, 9/16, 10.96, 2768, 8	16/16, 9/16, 16.61, 5408, 16

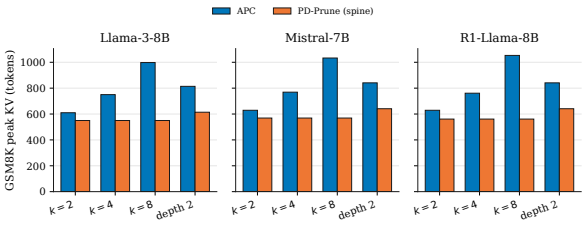


Figure 15: **GSM8K peak KV versus branching** (n=16, tp=2, D=512). The spine peak (PD-Prune and CoW-only) is invariant in k, while the APC peak grows with k.

tive residuals rejected) and $k/2$ at 0.45 (valid only). Relative T_{e2e} cuts under ESC+0.15 grow with k: 5%, 6%, 12%, and 17% at k=4, 8, 16, 32. Because SR and DPTS already terminate repetitive residuals during decoding, the same threshold yields smaller relative reductions (6% and 3% at k=16).

D.2 Prefill-only ESC bars

Table 15 isolates the prefill-side score: no decode-time pruning criterion is applied, and every admitted residual is decoded to D. Since $\text{Score}_{\text{pre}}$ takes three values on the mix (repetitive 0.02, malformed 0.22, valid ≈ 1), any $\tau_{\text{pre}} \in (0.02, 0.22]$ matches 0.15 and any $\tau_{\text{pre}} \in (0.22, 1)$ matches 0.45. At k=16, the τ_{pre}=0 row coincides in configuration with the ESC baseline of Table 14; committed accuracy across such runs ranges from 8/16 to 11/16.

D.3 Decode-only, prefill-only, and combined (k=16)

Table 16 factorizes the k=16 results. "Stopped" denotes the number of admitted residuals terminated by the pruner's decode-time criterion. Combined rows apply admission first and then the pruner criterion to the admitted set.

Table 15: Prefill-only admission (no decode-time cut; every admitted residual decodes to D).

k	τ_{pre}	admit	W	D^+	T_{e2e}	any	comm.
4	0.00	4	1320	32768	8.62	16/16	9/16
4	0.15	3	1018	24576	8.22	16/16	8/16
4	0.45	2	732	16384	7.90	14/16	9/16
8	0.00	8	2624	65536	10.90	16/16	8/16
8	0.15	6	2020	49152	10.25	16/16	10/16
8	0.45	4	1448	32768	8.66	15/16	8/16
16	0.00	16	5008	131072	16.48	16/16	9/16
16	0.15	12	3848	98304	14.45	16/16	10/16
16	0.45	8	2768	65536	10.95	16/16	8/16
32	0.00	32	10528	262144	32.78	16/16	9/16
32	0.15	24	7968	196608	27.14	16/16	9/16
32	0.45	16	5408	131072	16.61	16/16	9/16

D.4 Threshold-Based versus Top-1-Only Admission

Two policies share τ=0.45 but differ in how many residuals are admitted. *Score* admits every i with $s_i \geq \tau$ and index 0 ($k/2$ residuals on the mix). *Top-1 only* admits index 0 alone, the current highest-ranked candidate (W=414). Table 17 shows that Top-1-only admission lowers coverage (ESC $11/16$, Speculative Rejection $9/16$ at k=16), while score-based admission keeps $16/16$.

D.5 MATH-500: Blanket Rejection, Reject-Only Admission, and Probing

Table 18 compares the full fan-out, Blanket, reject-only admission, and truncating repetitive residuals after 128 decode tokens. Table 19 probes the band $[\tau_{\text{pre}}, \tau_{\text{dec}})$ for t_0 tokens and extends to D iff $s'_i \geq \tau_{\text{dec}}$=0.45.

Summary. (i) Blanket rejection of every s_i<τ_{dec}, i.e., aggressive prefill-side rejection, loses coverage: $20/32$ falls to $13/32$ at k=16. (ii) Rejecting only repetitive residuals costs at most one problem at $k \leq 8$ but reaches only $16/32$ at k=16. (iii) A t_0=128 probe increases coverage to $20/32$ at k=16 and cuts 104.10 s to 72.21 s (0.69×); at k=8 the shorter probe t_0=64 is the accuracy-preserving point ($16/32$ in 42.01 s, exceeding the full fan-out). (iv) Most uncertain residuals pass the decode-side refinement in this mix; probing repetitive residuals as well extends none of them and increases wall-clock time (74.44 s at k=16). (v) At k=16, probe extension recomputes evicted pages, so W exceeds that of the full fan-out although D^+ and wall-clock time decrease; retaining probe pages until the extension decision would eliminate this term.

D.6 Token Accounting Checks

Across Tables 14 and 15, the measured prefill satisfies the predictions of Proposition 1 for a warm trunk: at $k \leq 16$ the three pruners report identical W, and admission removes exactly the computed tokens of the rejected residuals. The control-plane connector table of Table 5 extends to wider fan-outs with the same 25% repetitive-only cut on a stock prefiller and the same alias+gate reduction to the miss of $\mathcal{K}$ (2240/4800/9920 tokens at k=8/16/32). Because residual lengths differ across classes, the reductions are not fixed fractions of W:

	k=4	k=8	k=16	k=32
W, no admission	1320	2624	5008	10528
W, τ_{pre}=0.15	1018 (0.77)	2020 (0.77)	3848 (0.77)	7968 (0.76)
W, τ_{pre}=0.45	732 (0.55)	1448 (0.55)	2768 (0.55)	5408 (0.51)

Repetitive residuals therefore account for about 23% of residual prefill and malformed residuals for a further 22%, although each class is a quarter of the residuals. Combined rows never exceed the prefill-only W at the same τ_{pre}.

E Extended Related Work

Inference-time search. Chain-of-thought prompting elicits intermediate steps [32]; Tree-of-Thoughts and Monte Carlo tree search expand, score, and backtrack over such steps [35, 39]; self-consistency samples complete traces and takes a majority [31]. Accuracy and token cost both grow with the number of traces. EquivPruner observes that textually distinct steps can be the same mathematical action and retains one representative per equivalence class [21]. PD-Prune neither changes the search procedure nor decides semantic equivalence; it decides, for the residuals a branching controller proposes, which are prefilled, transferred, and decoded.

Path pruning. ESC stops sampling once a window agrees [17]; Speculative Rejection halts low-reward partial generations at a horizon [26]; DPTS early-stops children after a mini-step [10]; STOP categorizes pruning signals as internal or external and learned or not, and trains an internal futility token [4]. These methods read generated tokens or compare written steps, so they act on the decode side of a PD deployment. PD-Prune adds a pre-kernel decision and a decode-side probe around them and leaves their decode-time cut unchanged (§C.4); a learned signal such as STOP's could replace $\text{Score}_{\text{dec}}$.

Table 16: Factorization at k=16 (GSM8K, Qwen3-8B). Decode-only: τ_{pre} absent, W=5008. Prefill-only: decode-time cut disabled, decode to D. Combined: both. Any-finished is $16/16$ on every row. Prefill-only and combined rows come from separate runs.

Factor	Setting	admit	stopped	W	D^+	T_{e2e}	comm.
Decode-only							
	ESC (D=512)	16	0	5008	131072	16.49	9/16
	SR (τ=256)	16	8	5008	98304	13.29	9/16
	DPTS (d=100)	16	8	5008	78336	11.87	10/16
Prefill-only							
	τ_{pre}=0.15	12	0	3848	98304	14.45	10/16
	τ_{pre}=0.45	8	0	2768	65536	10.95	8/16
Combined							
	ESC+0.15	12	0	3848	98304	14.45	9/16
	SR+0.15	12	4	3848	81920	12.53	9/16
	DPTS+0.15	12	4	3848	71936	11.57	10/16
	ESC+0.45	8	0	2768	65536	10.95	9/16
	SR+0.45	8	0	2768	65536	10.95	9/16
	DPTS+0.45	8	0	2768	65536	10.96	9/16

Table 17: Score-based versus Top-1-only admission at τ=0.45 (GSM8K, Qwen3-8B). Rows come from a separate run from Table 14.

k	Policy	Pruner	admit	W	T_{e2e}	any	comm.
16	base	ESC	16	5008	17.02	16/16	10/16
16	score	ESC	8	2768	10.91	16/16	10/16
16	Top-1 only	ESC	1	414	7.30	11/16	11/16
16	base	SR	16	5008	13.12	16/16	12/16
16	score	SR	8	2768	10.84	16/16	10/16
16	Top-1 only	SR	1	414	7.29	9/16	9/16
16	base	DPTS	16	5008	11.85	16/16	10/16
16	score	DPTS	8	2768	10.89	16/16	9/16
16	Top-1 only	DPTS	1	414	7.30	11/16	11/16
8	Top-1 only	ESC	1	414	7.28	11/16	11/16
8	score	ESC	4	1448	8.65	14/16	9/16
4	Top-1 only	ESC	1	414	7.29	10/16	10/16
4	score	ESC	2	732	7.88	14/16	11/16

Request-level serving. Orca introduced iteration-level scheduling [37]; vLLM pages the KV cache and hashes full blocks for APC [14]; SGLang shares prefixes through a radix tree [41]; SARATHI and Sarathi-Serve chunk prefills to co-schedule them with decode [3, 2]; DistServe and Splitwise disaggregate prefill and decode [42, 25]. Whether these engines share a trunk across concurrent siblings depends on the cache regime (§3.2); in all cases they serve every request they receive. PD-Prune uses their residency information as an input to admission and their connector as the channel whose volume admission reduces.

Program-, workflow-, and conversation-level scheduling. Autellix schedules by program-level attained service [23]; JITServe allocates just enough bandwidth under heterogeneous SLOs [40]; Hermes, ThunderAgent, and SAGA schedule at program or workflow granularity [22, 13, 11]; Parrot, Ayo, and Pythia exploit application structure [20, 27, 38]. ConServe disaggregates prefill and decode scheduling at conversation granularity based on observed agent behavior [9]. These systems decide when and where requests run; PD-Prune decides which sibling continuations become requests and at which stage a discarded one stops consuming resources.

KV retention, overlap, and copy-on-write. InferCept, Continuum, and MORI preserve or place context across tool calls [1, 16, 33]. Sutradhara overlaps tool execution with the prefill of a single tool-independent prefix [5]; SmoothAgent transforms long-horizon context; IdleSpec uses idle time for speculative planning to improve accuracy rather than KV reuse [24, 7]. ForkKV shares a base KV cache copy-on-write across multi-LoRA agents [29], which is closest to our context tree; PD-Prune applies copy-on-write to sibling continuations of one agent step and couples it with admission and spine retention. CachePrune edits KV neurons so that a model ignores instructions embedded in retrieved context [28]; it changes attention state within one request and does not decide which siblings are served.

Table 18: MATH-500 level ≥ 4, Qwen3-8B, n=32, D=1024. "Blanket" rejects every s_i<0.45. "Reject-only" rejects repetitive residuals only. "short probe@128" admits every residual but limits decoding of repetitive residuals to 128 tokens.

k	Setting	admit	W	D^+	T_{e2e}	any
4	full	4	2844	131072	25.34	12/32
4	blanket	2	1566	65536	19.11	6/32
4	reject-only	3	2189	98304	23.05	11/32
4	short probe@128	4	2844	102400	23.14	10/32
8	full	8	5592	262144	52.50	15/32
8	blanket	4	3036	131072	25.60	11/32
8	reject-only	6	4282	196608	41.53	17/32
8	short probe@128	8	5592	204800	42.26	15/32
16	full	16	12320	524288	104.10	20/32
16	blanket	8	6200	262144	52.05	13/32
16	reject-only	12	9332	393216	71.39	16/32
16	short probe@128	16	11856	409600	74.03	17/32

Table 19: Probe admission on MATH-500. Probe@t_0: repetitive residuals rejected; malformed residuals probed and extended iff $s'_i \geq 0.45$. Probe-rep.: repetitive residuals are probed as well (none is extended). At k=16, W includes the recomputation Δ_{ext} of evicted probe pages; it is about 19.4k tokens at t_0=128 and 11.2k at t_0=64 relative to reject-only (9332).

k	Setting	admit	W	D^+	T_{e2e}	any
4	probe@64	3	2460	98304	23.96	8/32
4	probe@128	3	2460	98304	24.70	9/32
4	probe-rep.@128	4	3115	102400	24.89	10/32
8	probe@64	6	4824	196608	42.01	16/32
8	probe@128	6	4824	196608	43.30	13/32
8	probe-rep.@128	8	6134	204800	43.32	15/32
16	probe@64	12	20544	393216	72.50	18/32
16	probe@128	12	28736	393216	72.21	20/32
16	probe-rep.@128	16	31260	409600	74.44	20/32

Speculative decoding. Speculative decoding and multi-head drafters propose *output* tokens and verify them against the target model [15, 6, 18]. The object PD-Prune may prefill speculatively is a future *prompt* prefix, such as a known tool-call suffix (Appendix B.3); speculatively prefilled pages are never sampled from, and (11) binds the prompt that actually arrives.